Haute École Arc Ingénierie
Filière : Informatique et systèmes de communication
Orientation: Ingénierie des données

Finding the hydrogen 21-cm signal

Inferring the reionization fraction from Gaussian and non-Gaussian data

Alex Houttuin

Présenté à : Emmanuel De Salis (Superviseur)
Expert : Nicolas Cerardi

Travail de Bachelor 2026
Présenté en vue de l'obtention du Bachelor of Science HES-SO



Neuchâtel, Suisse
24 juillet 2026

Déclaration

Ce travail a été réalisé dans le cadre d'une formation en ingénierie à la Haute École Arc à Neuchâtel en vue de l'obtention du Bachelor of Science HES-SO.

J'atteste avoir réalisé personnellement le présent travail dans le respect de la Directive de la Haute École Arc relative au plagiat du 6 février 2024.

Par ailleurs, j'indique avoir eu recours aux aides humaines et outils numériques suivants :

- **Aides humaines :** Emmanuel De Salis (discussions hebdomadaires et cadrage), Nicolas Cerardi (retours d'expertise).
- **Outils numériques :** Python (bibliothèque `sbi`, PyTorch, scikit-learn, matplotlib), MLflow (suivi des expérimentations), 21cmFAST (données simulées).
- **Outils d'IA générative :** Google Gemini (correction grammaticale, reformulation syntaxique et vérification du code LaTeX).

Date et signature : 24/07/2026 Alex Monthuin

Tableau de déclaration pour l'utilisation des IA génératives

	Corrections	**Reformulations**	**Synthèses**	**Explications**	**Inspirations**	**Citations**
Pas d'usage d'IA pour...	☐	☐	☒	☒	☒	☒
Utilisé l'IA pour...	☒	☒	☐	☐	☐	☐

Abstract

A major scientific goal of modern cosmology is understanding how the early universe went from dark to light during the Epoch of Reionization (EoR). To observe the hydrogen 21-cm radio emissions from this era, an international consortium is building the Square Kilometre Array Observatory (SKAO), which is a massive global radio telescope network expected to come online in the coming years. Processing the massive quantities of data this telescope will produce will require a strong mastery on machine learning techniques.

Traditional Bayesian methods require an explicit likelihood function, which is intractable for complex cosmological simulators, therefore, this Bachelor thesis explores Simulation-based Inference (SBI). From a computer science and data-processing perspective, we use deep neural networks to infer the timeline of cosmic reionization directly from mock data while providing uncertainty estimates.

We trained and evaluated six architectures for the SKA Science Data Challenge (SDC3b), comparing two distinct data types: 2D power spectra (measuring the variance in brightness temperature across spatial scales) and Betti numbers (counting discrete structures within the intergalactic medium). Our results demonstrate that while energy summaries provide a strong baseline accuracy, training on discrete structures allows the models to be up to 7 times more precise in some scenarios and epochs. Furthermore, combining both data types into a joint architecture achieves the best global accuracy, reaching $R^2 > 99.53\%$ for the joint ensemble.

Finally, we show that ensembling models provides a margin against overconfidence, which should help make our pipelines more robust when used with real telescope observations in the future.

Résumé

L'un des principaux objectifs scientifiques de la cosmologie moderne est de comprendre comment l'univers primitif est passé de l'obscurité à la lumière au cours de l'époque de la réionisation (EoR). Afin d'observer les émissions radio de l'hydrogène à 21 cm datant de cette époque, un consortium international construit actuellement le Square Kilometre Array Observatory (SKAO), un grand réseau mondial de radiotélescopes qui devrait entrer en service prochainement. Le traitement des volumes considérables de données que ce télescope produira nécessitera une bonne maîtrise des techniques de machine learning.

Les méthodes bayésiennes traditionnelles nécessitent une fonction de vraisemblance explicite, ce qui s'avère intraitable pour les simulateurs cosmologiques complexes, c'est pourquoi cette thèse de Bachelor explore l'inférence basée sur les simulations (SBI). D'un point de vue informatique et du traitement des données, nous utilisons du deep learning pour déduire la chronologie de la réionisation cosmique directement à partir de données simulées, tout en fournissant des estimations d'incertitude. Nous avons entraîné et évalué six architectures dans le cadre du SKA Science Data Challenge (SDC3b), en comparant deux types de données distincts : les spectres de puissance en 2D (mesurant la variance de la température de brillance à différentes échelles) et les nombres de Betti (comptant les structures discrètes au sein du milieu intergalactique). Nos résultats montrent que, si les résumés des spectres de puissance offrent une base solide, l'entraînement sur les nombres de Betti permet aux modèles d'atteindre une précision jusqu'à 7 fois supérieure dans certains scénarios et certaines époques. De plus, la combinaison des deux types de données au sein d'une architecture conjointe permet d'obtenir la meilleure précision globale, avec un $R^2 > 99.53\%$ pour l'ensemble conjoint.

Enfin, nous montrons que la méthode d'ensemble combinant plusieurs modèles offre une marge de sécurité contre la surconfiance, ce qui devrait contribuer à renforcer la robustesse de nos pipelines lorsqu'ils seront utilisés à l'avenir avec de véritables observations télescopiques.

Contents

1 Introduction **1**

2 State of the art **3**

3 Data analysis **5**
3.1 Data generation . . . 5
3.2 Neutral hydrogen fraction . . . 5
3.3 2 and 3-point statistics . . . 7

4 Conception **8**
4.1 Models . . . 8
4.1.1 SBI . . . 8
4.1.2 Ensemble posteriors . . . 9
4.1.3 Embedding networks . . . 9
4.1.4 Calibration in parameter space . . . 9
4.1.5 Calibration in observation space . . . 10
4.2 Metrics . . . 10
4.2.1 Regression metrics . . . 10
4.2.2 Scenario metrics . . . 10
4.2.3 Calibration metrics . . . 11

5 Implementation **12**
5.1 Iteration process . . . 12
5.2 Loading data . . . 12
5.3 Model implementation . . . 13
5.3.1 Initialization . . . 13
5.3.2 Gridsearching . . . 13
5.3.3 30 trials run . . . 14
5.3.4 Evaluation of the best model . . . 14
5.3.5 SBI trained on PS2D . . . 14
5.3.6 SBI trained on Betti numbers . . . 15
5.3.7 SBI trained on both PS2D and Betti numbers . . . 15
5.3.8 SBI Ensembles . . . 15

6 Results and discussion **17**
6.1 SBI PS2D . . . 17
6.1.1 Hyperparameters . . . 17
6.1.2 30 run trial TARP calibrations . . . 17
6.1.3 Final TARP calibration . . . 17
6.1.4 Posterior predictive check . . . 17
6.1.5 Reionization history . . . 19
6.1.6 Posteriors per scenario . . . 19
6.1.7 Pairplots per scenario . . . 20
6.1.8 Metrics . . . 20

6.2 SBI Betti . . . 21
6.2.1 Hyperparameters . . . 21
6.2.2 30 run trial TARP calibrations . . . 21
6.2.3 Final TARP calibration . . . 21
6.2.4 Posterior predictive check . . . 22
6.2.5 Reionization history . . . 23
6.2.6 Posteriors per scenario . . . 23
6.2.7 Pairplots per scenario . . . 24
6.2.8 Metrics . . . 25
6.3 SBI PS2D + Betti . . . 25
6.3.1 Hyperparameters . . . 25
6.3.2 30 run trial TARP calibrations . . . 25
6.3.3 Final TARP calibration . . . 25
6.3.4 Posterior predictive check . . . 25
6.3.5 Reionization history . . . 27
6.3.6 Posteriors per scenario . . . 27
6.3.7 Pairplots per scenario . . . 28
6.3.8 Metrics . . . 29
6.4 SBI 5-Ensemble PS2D . . . 29
6.4.1 30 run trial TARP calibrations . . . 29
6.4.2 Final TARP calibration . . . 29
6.4.3 Posterior predictive check . . . 29
6.4.4 Reionization history . . . 30
6.4.5 Posteriors per scenario . . . 30
6.4.6 Pairplots per scenario . . . 31
6.4.7 Metrics . . . 31
6.5 SBI 5-Ensemble Betti . . . 32
6.5.1 30 run trial TARP calibrations . . . 32
6.5.2 Final TARP calibration . . . 32
6.5.3 Posterior predictive check . . . 32
6.5.4 Reionization history . . . 33
6.5.5 Posteriors per scenario . . . 33
6.5.6 Pairplots per scenario . . . 34
6.5.7 Metrics . . . 34
6.6 SBI 5-Ensemble PS2D + Betti . . . 35
6.6.1 30 run trial TARP calibrations . . . 35
6.6.2 Final TARP calibration . . . 35
6.6.3 Posterior predictive check . . . 35
6.6.4 Reionization history . . . 36
6.6.5 Posteriors per scenario . . . 36
6.6.6 Pairplots per scenario . . . 37
6.6.7 Metrics . . . 37
6.7 Discussion . . . 38

7 Conclusion **40**
7.1 Summary of findings . . . 40
7.2 Recommendations for the SKA . . . 41
7.3 Mandate fulfilment . . . 41
7.4 Future perspectives . . . 41

List of Figures

3.1 Analysis of the parameter θ, showing the log-distribution, sorted percentiles, per-bin distributions and max/min outlier trajectories. . . . 6
3.2 Density plot showing the concentration of θ per redshift bin, showing the progression of a neutral IGM towards a ionized one. . . . 6
3.3 Side by side comparison of observations at a middle point of the EoR (index 8000, redshift bin 1). Left: PS2D, measures brightness temperature variance, lacks topological information Right: Betti numbers, tracks topology in the IGM. . . . 7

6.1 PS2D 30 run trial TARP calibrations . . . 18
6.2 PS2D TARP on the global test set . . . 19
6.3 PS2D PPC for low, middle and high scenarios respectively . . . 19
6.4 PS2D reionization history . . . 20
6.5 PS2D Posteriors per scenario . . . 20
6.6 PS2D pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization . . . 21
6.7 Betti 30 run trial TARP calibrations . . . 22
6.8 Betti TARP on the global test set . . . 23
6.9 Betti PPC for low, middle and high scenarios respectively . . . 23
6.10 Betti reionization history . . . 24
6.11 Betti Posteriors per scenario . . . 24
6.12 Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization . . . 24
6.13 PS2D + Betti 30 run trial TARP calibrations . . . 26
6.14 PS2D + Betti TARP on the global test set . . . 27
6.15 PS2D PPC for low, middle and high scenarios respectively . . . 27
6.16 PS2D + Betti reionization history . . . 28
6.17 PS2D + Betti Posteriors per scenario . . . 28
6.18 PS2D + Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization . . . 28
6.19 Ensemble PS2D TARP on the global test set . . . 29
6.20 Ensemble PS2D PPC for low, middle and high scenarios respectively . . . 30
6.21 Ensemble PS2D reionization history . . . 30
6.22 Ensemble PS2D Posteriors per scenario . . . 31
6.23 Ensemble PS2D pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization . . . 31
6.24 Ensemble PS2D pairplots of all 5 posteriors per scenario; left: fast reionization, middle: normal reionization, right: slow reionization . . . 31
6.25 Ensemble Betti TARP on the global test set . . . 32
6.26 Ensemble Betti PPC for low, middle and high scenarios respectively . . . 33
6.27 Ensemble Betti reionization history . . . 33
6.28 Ensemble Betti Posteriors per scenario . . . 34
6.29 Ensemble Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization . . . 34

6.30 Ensemble Betti pairplots of all 5 posteriors per scenario; left: fast reionization, middle: normal reionization, right: slow reionization 34
6.31 Ensemble PS2D + Betti TARP on the global test set 35
6.32 Ensemble PS2D + Betti PPC for low, middle and high scenarios respectively 36
6.33 Ensemble PS2D + Betti reionization history 36
6.34 Ensemble PS2D + Betti Posteriors per scenario 36
6.35 Ensemble PS2D + Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization 37
6.36 Ensemble PS2D + Betti pairplots of all 5 posteriors per scenario; left: fast reionization, middle: normal reionization, right: slow reionization 37

List of Tables

3.1 Statistics per-bin of θ 5

6.1 PS2D PPC metrics 19
6.2 PS2D scoring metrics 21
6.3 Betti PPC metrics 23
6.4 Betti scoring metrics 25
6.5 PS2D + Betti PPC metrics 27
6.6 PS2D + Betti scoring metrics 29
6.7 Ensemble PS2D PPC metrics 30
6.8 Ensemble PS2D scoring metrics 32
6.9 Ensemble Betti PPC metrics 33
6.10 Ensemble Betti scoring metrics 35
6.11 Ensemble PS2D + Betti PPC metrics 35
6.12 Ensemble PS2D + Betti scoring metrics 37
6.13 Overview of the model metrics 38

Chapter 1

Introduction

In the context of the Bachelor thesis in Informatique et systèmes de communications, we were given a mandate to infer the reionization fraction from the hydrogen 21-cm signal. This is not a thesis in the field of astrophysics; rather, this domain serves as the cosmological backdrop for a data-oriented machine learning project. To process and infer parameters from this data effectively, it is necessary to understand a minimum of the physical phenomena that generate it.

After the Big Bang and the cosmic dark ages that followed, the universe entered a period known as the Epoch of Reionization (EoR). During this time, the intergalactic medium (IGM) was filled with neutral hydrogen. When the first stars and galaxies formed, they began to emit high-energy ionizing radiation. This radiation started ionizing the surrounding neutral hydrogen. Because ionizing sources formed better in dense regions of matter, reionization proceeded in a patchwork fashion in the IGM, as it was merging bubbles around sources rather than uniformly everywhere. This structure is what makes the IGM a non-Gaussian environment. We can track this neutral hydrogen through the 21-cm signal, a radio emission produced by the spin-flip transition of neutral hydrogen atoms. As the universe ionized and neutral hydrogen disappeared, this 21-cm signal gradually faded away (for more information on this subject, refer to the books from S. Furlanetto et al. 2006; Mesinger 2020; Loeb and S. R. Furlanetto 2013).

To observe these signals, an international consortium of researchers and engineers formed the Square Kilometre Array Observatory (SKAO). The SKAO is a global effort to build the world's largest radio telescope network, with a combined collecting area of one square kilometre to fetch these radio emissions, in sites located in South Africa and Australia. The observatory is being constructed in stages, with early science operations beginning around 2027 to 2029 and scaling toward full operational capacity through the early 2030s.

Because the SKAO will produce complex and massive quantities of data, the consortium has set up a series of Science Data Challenges (SDC) to anticipate the preparation of data processing pipelines before the telescopes are operational. These challenges are designed to be data-driven, which allows computer science teams to participate and build models without requiring deep prior knowledge of astrophysical and cosmological notions (as it is in our case).

The third iteration of these challenges, SDC3, is divided into two parts: SDC3a focuses on filtering out foreground contamination to extract the 21-cm signal, while SDC3b focuses on inferring the physical properties of the EoR (specifically the reionization history) with uncertainty estimates. We focus our efforts only on the SDC3b challenge.

Because the EoR spans millions of years, the 21-cm signal is stretched by the expansion of the universe. This means that the frequency of the observed signal corresponds directly to a specific time in the universe's history (its redshift). Since we know the length of this signal, we can infer how long ago it was emitted by how much its wavelength has shifted to the red (the wavelength gets longer and loses energy). In the SDC3b dataset, the observations are divided into three distinct frequency ranges, or redshift bins:

- z_1: 151 MHz to 165.9 MHz
- z_2: 166 MHz to 180.9 MHz

- z_3: 181 MHz to 195.9 MHz

These ranges are divided in this manner to represent three distinct chronological phases of the reionization process: early, middle, and late reionization. By evaluating the signal across these three bins, models can track the evolution of the neutral hydrogen fraction x_{HI} rather than just extracting a single static average.

Traditionally, Bayesian parameter estimation in cosmology was done via algorithms like Markov chain Monte Carlo (MCMC). However, MCMC requires us to evaluate a likelihood function for every proposed parameter step. For the EoR, simulators like 21cmFAST are complex enough that there is no way to directly evaluate the probability of an observation given a set of parameters (the likelihood). The model can only be simulated, not inverted into a likelihood function. Because the likelihood cannot be explicitly evaluated (it is intractable), MCMC in this case is not applicable, and this limitation pushes us to explore methods such as Simulation-Based Inference (SBI), which we will detail later in Chapter 4.

Given this context, our mandate is to infer the reionization properties of the EoR from hydrogen 21-cm signals using mock datasets generated by simulators such as 21cmFAST. To achieve this, we established a structured scope of work (see Appendix B for the complete cahier des charges):

1. Literature phase, we learn and gain a better understanding of the SKAO's mission, the SDC3b challenge, and the astrophysics of the 21-cm signal to establish a baseline of knowledge to aid us in grounding our machine learning targets.

2. State of the art, we review and understand the models used by the top-scoring teams in the SDC3b challenge scored with the PS1 and PS2 datasets, while researching other architectures and frameworks used outside the data challenge.

3. Implementation phase, here we iteratively develop, train, and fine-tune a model to infer the EoR parameters from our dataset.

4. Deliverables, we document the findings in this report and prepare the defense presentation.

To support the implementation phase, we also set secondary objectives to ensure proper machine learning operations (MLOps) so that we may further learn how to set up production-ready pipelines. This includes implementing a reproducible training pipeline and utilizing MLflow to track all experiments, log hyperparameters, and evaluate model iterations rapidly.

The report is structured as follows: Chapter 2 reviews the current state of the art in simulation-based inference and examines other architectures used in the SDC3b data challenge and beyond. Chapter 3 provides explanations for the target θ parameter and the observations with numbers and plots. Chapter 4 details theoretical notions of SBI, of calibrating in two different spaces, and includes the metrics by which we evaluate the accuracy and calibration of our models. Chapter 5 provides the experimental methodology of our pipeline (how we iterate, load data, etc.). Chapter 6 presents our results, diagnostics and an analysis of our models. Finally, Chapter 7 summarizes our findings and discusses future perspectives and improvements.

Chapter 2

State of the art

A number of recent papers have inferred the EoR parameters from the hydrogen 21-cm signal observations, some in the context of the SKAO SDC3b data challenge. All of these have explored the use of simulation-based inference (SBI) and have employed a large range of techniques and methods to achieve satisfying results.

The first paper to be read was a pre-print from Cerardi et al. 2025. The research team uses the same mock data generated by 21cmFAST that filters out extreme scenarios, and evaluates the constraining power of higher-order statistics (bispectrum and Betti numbers) and combines them with the 2D power spectrum (PS2D, see Section 3.3) using figures of merit computed over the full test set instead of a handful of examples. The key finding is that PS2D + Betti improves the figure of merit over PS2D alone across the full range of the reionization history, and this shows that the topological information is mostly not carried by PS2D (a two-point statistic). The bispectrum improves constraints in low to mid mean x_{HI} (≤ 0.6) but degrades the figure of merit in a late reionization scenario. The posteriors are validated using a method called TARP and 30 independent trial runs. As will be seen later in this report, a similar TARP-based and multi-run pipeline is employed (see Section 5.3.3).

The second paper is from De Salis et al. 2026. Instead of focusing on a single model, the team explored a larger set of deep learning architectures on the same PS2D dataset but augmented and with the average noise subtracted. These architectures are SE-CNN (Squeeze-and-Excite), MLP-Mixer, GLOW normalizing flows, frequency-aware CNN, MiniViT and SBI, including ensembles for SE-CNN. The metrics used to score were R^2 and RMSE, and they evaluated overall and per frequency bin. The SE-CNN Ensemble-10 achieved the best overall R^2 at 0.986 and RMSE at 0.032, while SBI ranked sixth. Per-bin, the MLP-mixer had slightly better scores for the two first frequency ranges, and the SE-CNN Ensemble-10 could infer better at the last bin. Based on this work and discussions with De Salis (also the supervisor of this project), the use of Kolmogorov-Arnold networks was excluded from consideration as other groups had not much success implementing a model with satisfying results.

Thirdly, we read a paper from Pietschke et al. 2025, who proposes an SBI framework called EoRFlow, that reconstructs the mean hydrogen fraction based on redshift ranges piece by piece from PS2D data. The model produces accurate and well calibrated posteriors for both typical and atypical reionization histories. Realistic noise in observations does not degrade stability, one scenario shows it even improves it, so it can be seen as an implicit regularizer. A noise assumption akin to the foreground emissions wedge from the SDC3a challenge broadens the posteriors but they still remain constrained enough to be useful. Inference is also fast (with about 5 seconds for each posterior) and the authors argue the framework is well-suited for the large volume that will be output when the SKA comes online.

The fourth paper from Cooper et al. 2026 compares two strategies for training an SBI pipeline to infer the parameters and compute the reionization history from PS2D. They train directly on noisy 21cmFAST simulations and on noisy samples sampled from a PS2D emulator. Those two approaches can compute parameters effectively. However, calibration tests are better when training on simulations directly. This is due to the emulator not correctly carrying enough sample variance according to the authors. Despite this, when the parameter posteriors are computed through a second emulator trained on the reionization history, both pipelines are able to get satisfying results on recovering it.

This suggests the emulator way of training is a viable option for future inference, despite having some weaker performance.

For our fifth paper, we read Zhao et al. 2024. Their team compressed raw 3D tomographic 21-cm lightcone images via a data transformation method called wavelet scattering transform (WST), before passing the newly generated input data to DELFI (a variant of SBI's density estimation). They call this approach 3D ScatterNet, and it outperforms both a trained 3D CNN compressor and a power spectrum DELFI pipeline. It also uses three signal types: pure signal, SKA noise and SKA noise with residual foreground. Because WST uses fixed and untrained kernels, it is also more efficient and robust compared to a CNN compressor which needs to be trained, though its use requires raw lightcone images, and they were not readily available in the scope of this project.

As the available time and computing was limited in the scope of this project, we decided to consolidate our efforts on a single architecture rather than spreading ourselves thin training many different architectures; and to use PS2D and Betti for our chosen data types. As such, we envisioned training 6 models, SBI and SBI 5-ensemble for each of PS2D, Betti and PS2D + Betti.

Chapter 3

Data analysis

3.1 Data generation

The simulated observations and their corresponding parameters were generated using the 21cmFAST algorithm, a cosmological simulator. To ensure our models stay consistent with the cosmological reality, extreme scenarios were discarded prior to our analysis (for a discussion on the simulation parameters and filtering criteria, refer to Cerardi et al. 2025).

Our primary target parameter θ represents the neutral hydrogen fraction x_{HI} across three distinct redshift bands. To account for telescope noise and observational uncertainty, each parameter set in our dataset is paired with 20 independent noise realizations. The following list details the shapes of parameter and observation tensors:

- Parameter θ has a true shape of `(3,)`, for all 20 noise realizations to align with the observation tensor. The dataset stores it as `(20, 3)` for convenience's sake, though all 20 rows are identical.
- PS2D observations have a shape of `(20, 3, 10, 10)`. This represents 20 noise realizations across 3 redshift bins within a 10×10 $(k_\perp, k_\parallel)$ grid.
- Betti observations have a shape of `(20, 3, 3, 21)`. This represents 20 noise realizations across 3 redshift bins for 3 distinct Betti numbers (b_0, b_1, b_2) over 21 density thresholds.

3.2 Neutral hydrogen fraction

The complete dataset consists of 15'951 samples for the three redshift bins. Figure 3.1 presents an exploratory data analysis on the θ parameter.

The sorted mean θ values across the three redshift bins (top-right) let us choose scenarios based on the dataset's percentiles. The distribution has an absolute minimum of 0.00000 and maximum of 0.95377, with the percentiles defined as follows:

- 16th percentile (fast reionization): 0.31692
- 50th percentile (average reionization): 0.60760
- 84th percentile (slow reionization): 0.83299

Table 3.1: Statistics per-bin of θ

Statistic	z_1	z_2	z_3
Minimum	0	0	0
Maximum	0.974	0.956	0.941
Lower boundary samples ≤ 0.01	12	124	1038
Upper boundary samples ≥ 0.99	0	0	0

Figure 3.1: Analysis of the parameter θ, showing the log-distribution, sorted percentiles, per-bin distributions and max/min outlier trajectories.

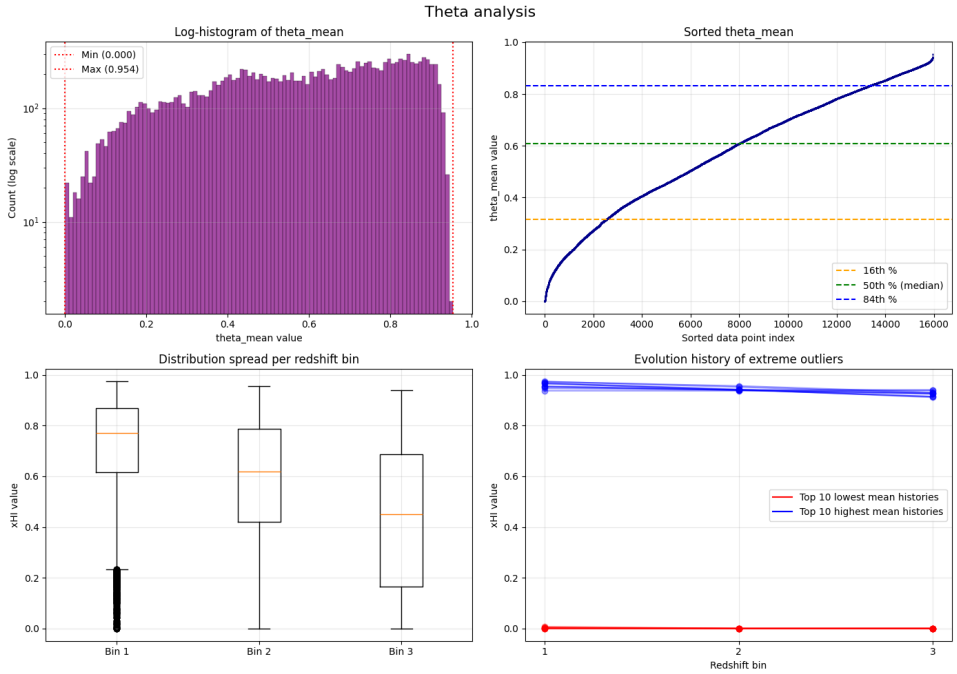
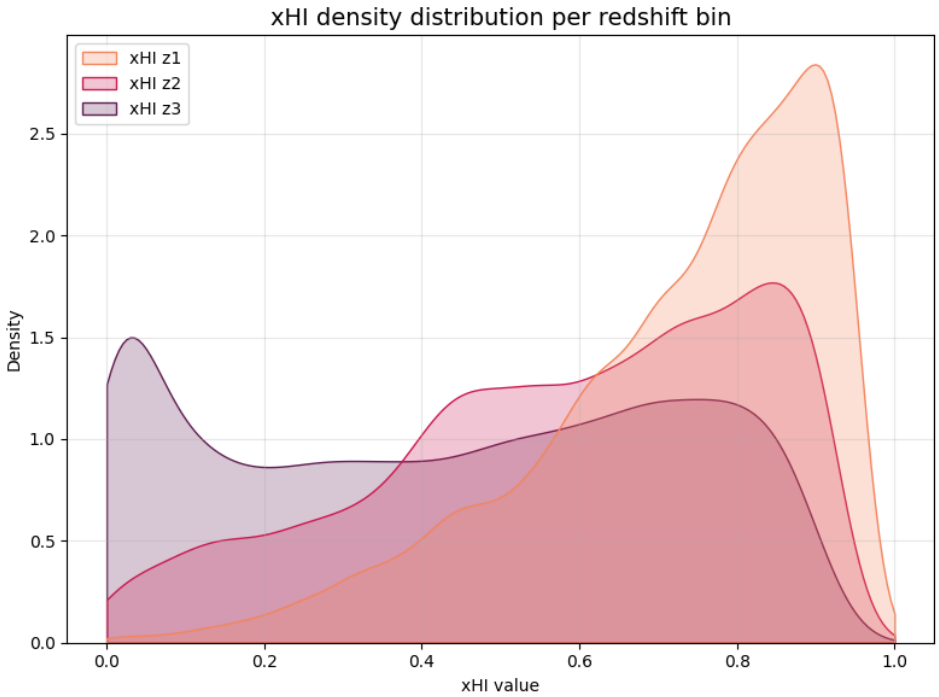


Table 3.1 numerically shows us the evolution of the EoR. As time progresses, the number of samples being ionized completely ($x_{\text{HI}} \leq 0.01$) increases drastically from the earliest redshift bin z_1 to the latest z_3. Figure 3.2 shows a density plot per redshift bin which also highlights this evolution. The earliest epoch z_1 shows us that the concentration skews close to 1, meaning the IGM was largely neutral at this stage. z_2 indicates a wider distribution, especially between 0.4 and 0.9 and z_3 has a heavy skew near 0, and a flatter distribution with a smaller mode, between 0.2 and 0.8.

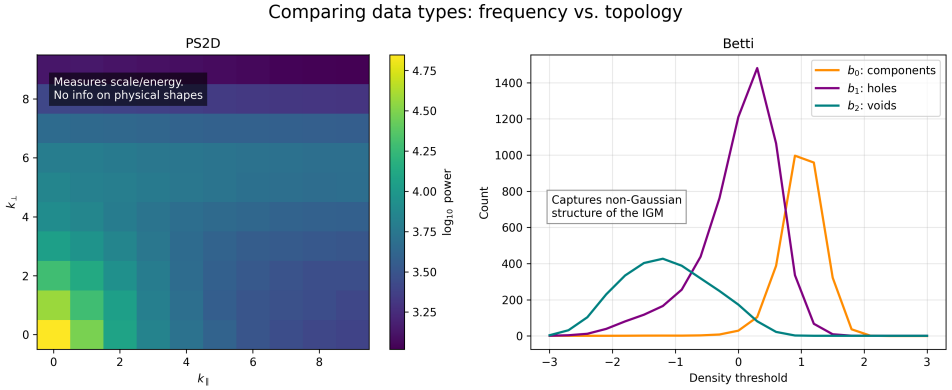
Figure 3.2: Density plot showing the concentration of θ per redshift bin, showing the progression of a neutral IGM towards a ionized one.



3.3 2 and 3-point statistics

To infer the θ parameter with success during the EoR, our machine learning pipeline will pass as input data summaries that hold pertinent information on the IGM. As plotted in Figure 3.3, our two chosen data types offer different insights.

Figure 3.3: Side by side comparison of observations at a middle point of the EoR (index 8000, redshift bin 1). Left: PS2D, measures brightness temperature variance, lacks topological information Right: Betti numbers, tracks topology in the IGM.



The 2D power spectrum (**PS2D**, left panel) measures the variance in brightness temperature differences along the parallel and perpendicular lines of sight. In this way, we obtain slices that amount to a hologram of the EoR, containing information on the overall energy from the ionizing hydrogen signal. However, this 2-point statistic integrates out spatial patterns present in the IGM. The power spectrum is cylindrically projected into $(k_\perp, k_\parallel)$ Fourier mode bins (this is measured in spatial frequencies of Mpc^{-1}) to average down noise, mitigate wedge foreground emissions, and reduce dimensionality. PS2D measures temperature variance in units of mK^2 and the values span a large dynamic range, scaling from $\sim 4.7 \times 10^2$ to amplitudes up to $8.5 \times 10^6\,\text{mK}^2$ during reionization.

To gain information on the topology, we use **Betti** numbers (right panel) to track the structural evolution of the IGM across density thresholds. Here, b_0 counts ionized regions, b_1 interconnecting tunnels and loops, and b_2 enclosed neutral voids. Unlike the continuous intensity variance measured by PS2D, Betti numbers represent discrete feature counts ranging between 0 and $\sim 4'900$. In a standard Gaussian random field, these curves would form a symmetrical distribution; however, our observations show some asymmetry where the peak count of connected ionized regions ($b_0 \approx 1000$) exceeds the peak count of enclosed voids ($b_2 \approx 420$). This asymmetry shows the non-Gaussian nature of the IGM during reionization (For more information on PS2D and Betti numbers: Mesinger 2020).

Chapter 4

Conception

4.1 Models

To infer the neutral hydrogen fraction x_{HI} from 21-cm observations, there are various models and architectural frameworks that can be deployed, as seen in our State of the Art chapter. We have centered our efforts on simulation-based inference (SBI).

4.1.1 SBI

Simulation-based inference (SBI) is a set of methods to learn parameters knowing a prior $p(\theta)$ and simulated data. Equation 4.1 defines the posterior we would like to predict, but the right-hand side of the equation requires the likelihood $p(x_o|\theta)$ which is intractable for simulators such as 21cmFAST. SBI will not evaluate the equation directly, but trains a neural network to approximate $p(\theta|x_o)$ from simulated (θ, x) pairs alone, that way it does not compute the likelihood explicitly.

$$p(\theta|x_o) = \frac{p(x_o|\theta)p(\theta)}{p(x_o)} \tag{4.1}$$

SBI lets us employ neural networks to approximate the posterior or the likelihood, and in this case we are looking to use the Neural Posterior Estimation (NPE), a trainer for a normalizing flow. A normalizing flow is a neural network that learns an invertible mapping between a base distribution (like a Gaussian distribution) and the target posterior. The mapping is bijective so the flow can generate samples from the posterior and evaluate the probability densities for a given θ parameter set (which is used for calibration diagnostics). NPE trains the flow to transform samples from the base distribution into samples from $p(\theta|x)$, which is conditioned on the observation.

Step by step, the NPE pipeline progresses like this:

- First, we define the prior $p(\theta)$, which is used to establish a plausible boundary range for the parameter.
- A number of θ parameters are sampled from this prior randomly, denoted as $\theta_i \sim p(\theta)$.
- For every sampled θ_i, we generate an observation x_i calculated by a simulator. This gives us training pairs (θ_i, x_i) in addition to the prior.
- A neural density estimator (the normalizing flow) is instantiated together with an embedding network for compressing the observations x and these are combined into a single object.
- The density estimator, together with the prior $p(\theta)$ defines an inference object
- The simulated pairs (θ_i, x_i) are appended to this inference object which then trains the density estimator by minimizing the negative log-likelihood of the true parameters under the estimated posterior. This is a single step and it will redo this step for many batches until the loss, measured on the validation set, converges.

- Once the training is finished, the density estimator can be wrapped into a posterior object, which can be sampled with a new observation x_o to produce our predicted $\hat{p}(\theta|x_o)$. This allows us to infer θ values together with uncertainty.

4.1.2 Ensemble posteriors

Training an individual SBI model is subject to the random initialization of weights and the random splitting of the train and validation sets. We will see during our implementation we have an empiric way of dealing with this. To further stabilize inference of the posterior and mitigate individual artefacts, we can select multiple well-calibrated normalizing flows to construct a posterior ensemble. During inference, the ensemble will average the predicted probability densities of its posteriors, which gives a mixture distribution that, in theory, is more robust, less prone to artefacting and less sensitive to random biases of individual posteriors.

A practical consequence happens during rejection sampling near bounded priors. When a true parameter is located close to a prior boundary (such as when it is close to 0 or 1), an individual flow can place a portion of the mass of probability outside the prior range. When sampling a bounded prior, the `sbi` package will apply rejection sampling to reject any samples from being drawn outside the boundaries. When combining multiple posteriors to make a mixture distribution, the resulting posterior is smoothed out and may spread the mass more outside the boundaries. If there is too much mass outside the valid bounds, the rejection sampling acceptance rate will drop to near zero, which can make sampling intractable. To prevent any failures for when sampling, we widen the prior boundaries to the full interval $[0, 1]$. Since this is only marginally wider than the empirical range of the training data (see Chapter 3), the effect on inference should be minimal.

4.1.3 Embedding networks

For both data types (PS2D and Betti), the observations are multi-dimensional and this can be difficult for the model to train, as it suffers from the curse of dimensionality and can become too computationally expensive. To compress these observations into summary statistics without losing critical information, we employ embedding networks that reduce the dimensionality to a given dimension.

There are multiple ways to train these networks, and we explore two of them:

1. The embedding network is instantiated directly inside the SBI architecture, in which case the weights are optimized simultaneously with the normalizing flow's parameters. They both act on the same loss function, so the network learns to compress the inputs to maximize the posterior inference accuracy. This method is used for the Betti dataset.

2. We independently pre-train a supervised neural network (such as a ResNet) using a standard regression criterion (like the mean squared error) to directly predict the parameter θ from the observation x. Once it has converged, the weights are frozen. During the SBI training, this network works as a feature extractor so that the input observations get a corresponding summary statistics, which are then fed into the normalizing flow. We use this method for the PS2D dataset.

Neither approach is better than the other, although the second one gives us more control over the representation learning process. A more complex version of this is called the variational mutual information maximization (VMIM), which can be read more about in Cerardi et al. 2025.

4.1.4 Calibration in parameter space

A model with a low validation loss after convergence is not guaranteed to be reliable. An uncalibrated posterior may be over/underconfident and have consequential misspecification, meaning the posterior may be too constrained or too wide with its uncertainty bounds. To evaluate the parameter space calibration for a given model, we use a method called the Test of Accuracy with Random Points (TARP).

TARP evaluates whether a model's predicted confidence intervals are truthful across an entire validation set. If a posterior distribution claims to be 95% credible, the true ground-truth parameter

should fall inside that credibility region 95% of the time for a large amount of test observations. For each test observation, TARP checks whether the true θ is inside the predicted α% credible region, for every credibility level between 0% to 100%, and then it aggregates this for the entire test set.

4.1.5 Calibration in observation space

In addition to this, we complement the TARP diagnostic with posterior predictive checks (PPC) to validate the model's reliability in the observation space. The PPC will verify whether the inferred parameters can be used to map back into realistic observations.

Normally, θ_{pred} parameters are sampled from the predicted posterior distribution given an observation x_o and passed to a simulator to generate back predicted observations x_{pred}. We can then directly compare these predicted observations with the actual observation. In our case, we do not have access to a simulator at inference time, we describe a method using k nearest neighbor as a proxy in Section 4.2.3.

4.2 Metrics

4.2.1 Regression metrics

For each test observation, we sample a set of parameter samples from the NPE-predicted posterior and compute the expected value (mean $\hat{\theta}$) to serve as our point prediction. This prediction is then evaluated against the ground-truth parameter θ using three regression metrics.

The coefficient of determination R^2 (see Equation 4.2) will be our primary score to compare our models' proportion of variance in neutral hydrogen fraction x_{HI}. It is a standardized metric allowing us to directly compare the performance of our models against existing literature (such as the models by De Salis et al. 2026).

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(\theta_i - \hat{\theta}_i)^2}{\sum_{i=1}^{N}(\theta_i - \bar{\theta})^2} \tag{4.2}$$

We evaluate both overall and per redshift bin, as the signal of x_{HI} is predominant in the early stage but also contaminated by the foreground emissions, both of these become weaker while the telescope noise might become relatively stronger. This way we can evaluate to see the evolution of the metrics as time progresses in the EoR to see if, despite interference, it may still predict with success.

In addition to the R^2 score, we also compute the root mean square error (RMSE) and the mean absolute error (MAE) (Equations 4.3 and 4.4). Because RMSE squares before averaging, it amplifies prediction errors caused by large discrepancies, whereas MAE is linear with the magnitude of the error. We can analyze the ratio and divergence between RMSE and MAE to determine the distribution in error. If RMSE is far greater than MAE, then the model is making severe outlier predictions for test samples.

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\theta_i - \hat{\theta}_i)^2} \tag{4.3}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\theta_i - \hat{\theta}_i| \tag{4.4}$$

4.2.2 Scenario metrics

We will be basing the model's predictive performance against the mean ground truth of x_{HI} within three of the representative scenarios:

- Average reionization (median, 50th percentile), this is the median ground truth, and the most likely trajectory where reionization progresses at an average rate.

- Fast reionization (-1σ, 16th percentile), this shows a possible reionization where the presence of x_{HI} decreased rapidly and starts **low**.
- Slow reionization (+1σ, 84th percentile), this represents a scenario where x_{HI} remains **high**, and decreases at a steady and low pace.

The reionization scenarios are within one standard deviation (in percentiles, the 16th, 50th and 84th) of a Gaussian distribution, as to show a possible reality within reasonable (non-extreme) boundaries and largely avoid noise or outliers.

4.2.3 Calibration metrics

For the TARP, we evaluate its p-value per model. The p-value tests the null hypothesis that the predicted posterior distribution is well calibrated, which we evaluate against a standard type-I error threshold of $\alpha = 0.05$. To get this p-value, we first compute the expected coverage probability (ECP) curve for each credibility levels from 0% to 100%. An ideal posterior will have an ECP curve that is precisely in line with the diagonal ($y = x$). The computed p-value will tell us the statistical significance of any deviations from the diagonal. As sampling the posterior is stochastic, we compute 20 different realizations and average the p-value of each of them to get the mean p-value, to ensure the model is indeed well calibrated through a robust test.

For the PPC, we do not have access to the simulator during inference, so we employ the k nearest neighbors (KNN) to identify the nearest parameters in the test parameter grid for each sampled θ_{pred}. We average the corresponding 20 observational noise realizations of these nearest neighbors to compute a proxy predicted observation, x_{pp}. We evaluate the KNN-assisted PPC using three complementary metrics:

- The mean absolute z-score, this measures how many standard deviations away the observed data is from the ground truth observation. A z-score of 1 means that, on average across the frequency bins, our predicted proxy observations x_{pp} differ from the true test observations by one standard deviation.
- The mean fractional residual, which is the relative error percentage between the true test observation and the predicted proxy one.
- And the mean nearest neighbor distance, this measures the Euclidean distance in the parameter space between the posterior samples θ_{pred} and the closest point in the test set. The closer the distance is, the more confident we can be that the test observation x_{pp} acts as a proxy for what the simulator would output.

In addition to numerical metrics, we plot scatter diagrams in observation space. In each PPC scatter plot, a single point corresponds to one dimension of the flattened observation vector (for example, one pixel of the PS2D or one Betti curve value at a given threshold), not one posterior sample. The horizontal axis gives mean(x_{pp}), the value of that specific observable dimension averaged for all KNN proxy observations for all sampled θ_{pred}. The vertical axis gives x_{obs}, the true value of that same dimension in the test observation. A point on the diagonal indicates that the model's posterior samples map back (on average) to the correct value for that observable feature. Points above the diagonal indicate that the true signal is stronger than predicted (underprediction), and vice versa.

All observations are plotted unscaled, meaning that their physical units stay the same. PS2D values can reach up to high amounts ($\sim 8.5 \times 10^6\,\text{mK}^2$) (measuring the 21-cm brightness temperature variance), whereas Betti curves will reach up to ~ 5000 (counting the topological features).

The PPC will not be used in determining a best model, rather it will show us, after training, additional information which may explain any oddities (such as the shape of the TARP, per-bin metrics which increase or decrease as the EoR advances, etc.).

Chapter 5

Implementation

During our implementation, rather than picking and matching pieces from various sources, we decided to simply follow the `sbi` Python package's documentation (Tejero-Cantero et al. 2026) as their tips, tricks and recommendations are plentiful and well-founded.

5.1 Iteration process

To speed up iteration and maintain a record of our training configurations, we integrated MLflow into our pipeline. MLflow is an open-source platform designed to manage the machine learning lifecycle. It provides a dashboard that automatically tracks hyperparameters, records training metrics in real-time, and archives artefacts (such as model weights, loss curves, evaluation plots, etc.). This tool is important for getting a fast overview on training our models.

We implement an adapter that connects `sbi`'s internal training callback (the tracker) to MLflow's logging API, so that the validation loss is automatically logged at every epoch of every trial, without modifying `sbi`'s training loop directly.

If a trained model's posterior for a given x_o represents the same distribution of θ shown in Figure 3.2 rather than varying across different observations, then it indicates a failure in constructing the posterior. The model has learned to ignore x and is outputting something close to the marginal distribution of $p(\theta)$. We encountered this in practice during development, where a mismatch between a frozen embedding network's expected input (raw observations) and the flow's automatic input normalization (to the mean and the variance) caused this failure; disabling normalization for pretrained embeddings solved this.

5.2 Loading data

Both our data types share the y target (the parameter θ) and both possess the 20 realizations. It is possible then to develop a generic dataloader so that we may retrieve our data without having a transformation pipeline bespoke for each data type.

There are three ways to load observations regarding their 20 noise realizations:

1. We can average all 20 noise realizations to get a $\bar{x}_o$ for a given parameter, but in this case we lose the variance, although we gain in speed.

2. We can pick a noise realization randomly, but every θ will be given an artefacted noisy observation which never quite matches reality. It is similar to having 20 blurry pictures, at best we may get a slightly clearer, but still blurry picture, but at worse we may get an even blurrier picture. This does have some merit if we would like more uncertainty, at the expense of R^2 scores. This could also lead to another issue where the TARP could report a good calibration since it evaluates against the same single-realization-per-θ likelihood the model is trained on, but it would not capture whether the model generalizes correctly to the full range of the realistic noise.

3. We can use all 20 noise realizations and treat them as paired samples $(\theta, x_{o,1}), \ldots, (\theta, x_{o,20})$, we preserve the variance of the signal but we risk reverse engineering the simulator's specific noise generation algorithm, which could cause the model to overfit to the artefacts of 21cmFAST rather than actual data, though no real observations are available until the SKA comes online. This technique is also the slowest.

We chose to continue with the 3rd option since providing the flow with all 20 noise realizations per parameter gives it a more accurate picture of the likelihood $p(x|\theta)$, which we found in practice to improve both predictive accuracy and calibration compared to using a single random realization.

5.3 Model implementation

Since our data types' observations are different from one another in nature and shape, they require a specific architecture, in that they will require different embedding networks to jointly train with the normalizing flow of the SBI.

5.3.1 Initialization

For a given data type, a DataLoader will load the correct dataset's observations and parameters. The prior is calculated based on the θ parameter's low and high value for the three redshift bins.

To ensure that we have no data leakage, the data is split into an 80/20 train-validation split and an isolated global test set before any transformation. The global test set is only used during the final evaluation of the model, so that it can benchmark the model's generalization to unseen data.

5.3.2 Gridsearching

Gridsearching is done in addition to k-fold cross validation to ensure that the given validation set does not differ too much from the training set. We explore these three hyperparameters, in order to minimize the negative log validation loss:

- Summary dimensions, (3, 16, 32). 3 corresponds to compressing the observation down to the dimensions of the θ parameter, while 16 and 32 test whether a higher dimensional summary statistic could improve the posterior's quality by having more information than a point-estimate.
- number of bins, (10, 32). This will control the granularity of the NSF spline transforms, more bins will allow the flow to represent more complex shapes, at the cost of potentially overfitting.
- learning rate, (0.0001, 0.0005). We test whether a faster (5×10^{-4}) or slower (1×10^{-4}) convergence rate will give a better validation loss score.

By iterating, we have also found that these parameters consistently outperformed others, and as such are removed from the gridsearching pool:

- NPE model NSF, and not MAF. We iterated and found the NSF model to be better, and SBI docs recommended to use it, although there is a risk of overfitting.
- Hidden features at 128 and number of transforms at 8, as we found these to give satisfactory results and did not want to explore too many combinations as it would be too computationally expensive.

In total, we have 12 combinations of hyperparameters, and with 3 K-folds, we need to train a given model 36 times to find the best hyperparameters. The combination with the best negative validation loss mean for the 3 folds will then be saved in a JSON file to be reused later. Cross-validation is used so that the generalization capabilities of a given hyperparameter combination is present, and not due to an artefact of a particular train/validation split. We average the three independent folds to give us a reliable ranking between combinations.

5.3.3 30 trials run

Based on the best hyperparameters, we will train a model 30 times with different random splitting, determined by a pseudorandom number generator with the trial's run number as the input. The reason we train many times is that SBI's training can vary quite a bit in validation loss and calibration due to its stochastic nature, and as such we need to find the best model empirically.

For the 30 trials run, we will select the model with the best mean p-value of their 20 TARP realizations, to ensure we have the most calibrated model. This is done because TARP itself is noisy as we draw samples from the posterior randomly, so multiple realizations are needed to average out the noise.

The best calibrated model is preferred to the model with the best validation loss, as we want a model that is well calibrated (not under, or overconfident) even if we have to trade-off a more constrained posterior, in addition this may be more preferable as constraining a model too much based on simulated data may not work as well when inferring real-world cosmological data. Because the hyperparameter combination is itself reasonably good in terms of validation loss during gridsearching, we have a solid baseline for every one of the 30 trials. Selecting them by TARP p-value will not heavily penalize the validation loss of the trained model, and the only trade off is therefore the calibration quality. This ensures we don't have a well calibrated model chosen on a poorly configured architecture.

5.3.4 Evaluation of the best model

To verify the predictive accuracy and reliability of the selected model, we evaluate on the global test set, which is held-out from the training and hyperparameter selection. The metrics and plots are saved to the filesystem and logged as artefacts in MLflow, alongside the best posterior and density estimator.

First we assess the predictive precision in parameter space. For each global test observation, we sample 1000 realizations from the posterior and compute the mean as a point prediction, which we compare against the ground truth using R^2, RMSE, and MAE, both overall and per redshift bin.

We then segment the test set into three scenarios (fast, average and slow reionization) and graph, for each, the reionization history plot, the pairplot of the sampled posterior and the posterior distributions. This is to give a view of both parameter and observation space for each scenario and redshift bin.

Next, we recompute the TARP calibration with 20 realizations over the global test set, to confirm that the model is still well calibrated (versus the validation TARP) when generalizing with unseen observations.

Finally we compute the PPC metrics and diagnostic plot described in Section 4.2.3, for the same three scenarios to assess whether the model's inferred parameters can map back to observations realistically.

5.3.5 SBI trained on PS2D

We train a ResNet-18 architecture on the PS2D observations, modified in two ways to suit our small 10×10 inputs: the initial convolutional layer's kernel size, stride and padding are reduced (kernel size 3, stride 1, padding 1) to avoid the downsampling originally meant for larger images, and the subsequent max-pooling layer is replaced with an identity mapping (basically a passthrough) since pooling at this small of a resolution would discard too much information. At the output layer, the network originally had a classification output, which we replaced with a linear layer that maps to 64 features. We add onto that a dropout at 10%, a ReLU function and then a final layer that reduces down to the 3 output values corresponding to our predicted $\hat{\theta}$. We train the model using MSE as our loss criterion.

This trained network is then used as our embedding network. We freeze all its weights so that its parameters no longer update during the flow's training, and we set it to evaluation mode so that the neuron dropout is deactivated. This is because the normalizing flow requires a deterministic function for mapping from the observation x to its embedding at both training and inference time.

Because we pretrain the embedding network ourselves on raw observations, we need to disable `sbi`'s automatic input normalization (`z_score_x="none"`) for this data type. Leaving the default

active would standardize inputs to the dataset's mean and variance before passing them to the ResNet, which was not trained on these inputs, leading to the `sbi` to largely ignore x and output posteriors close to the prior.

The ResNet's weight initialization is governed by PyTorch's global random number generator, which is seeded per trial with the trial's run number. This means, in theory, that the pipeline is reproducible, but we have seen that a convolutional algorithm on CUDA can possess non-determinism, so this is not guaranteed unfortunately PyTorch Contributors 2026.

5.3.6 SBI trained on Betti numbers

Compressing Betti numbers can be done with an MLP architecture, instead of a CNN. Betti curves do not have a 2D structure such as the PS2D, it is more of a 1D structure over 21 density thresholds per redshift bin and Betti number type (b_0, b_1, b_2). Because this network relative to the ResNet is shallower, we can jointly train it with the normalizing flow without compromising information and computational speed as it can receive a strong enough training signal from the normalizing flow's loss, rather than having a separate pretraining.

The MLP is structured as such: we first flatten the observations' shape from `(batch, 3, 3, 21)` down to `(batch, 189)`. Our first layer to receive the forward x will output to 128 neurons, followed by a layer normalization, a leaky ReLU, and a dropout at also 10%. The next layer will connect those neurons to 64 outputs, followed by a leaky ReLU as well. Our output layer will get these 64 neurons in and compress down to the given summary dimension.

We use leaky ReLU instead of the standard ReLU like in the output layer of ResNet to avoid the risk of dead neurons since Betti-derived features can potentially output negative values and a standard ReLU would zero the gradient entirely.

5.3.7 SBI trained on both PS2D and Betti numbers

In case of the joint training of PS2D + Betti, the data differ in terms of structure (as explored in Figure 3.3) so combining them requires preserving as much as possible their unique information before fusing them into a single summary statistic. A joint dataset wraps the two individual datasets, where for a given index, the shared θ parameter is loaded once, while each data type's 20 noise realizations are flattened (PS2D from `(batch, 20, 3, 10, 10)` to `(batch · 20, 300)` with $3 \cdot 10 \cdot 10 = 300$ and Betti from `(batch, 20, 3, 3, 21)` to `(batch · 20, 189)` with $3 \cdot 3 \cdot 21 = 189$) and then concatenated into a `(batch · 20, 489)` tensor per parameter.

Internally, the joint embedding network reverses the concatenation. It receives as input the flattened 489 tensor, slices the first 300 features and reshapes them back to `(3, 10, 10)` before passing to the frozen pretrained PS2D ResNet, and slices the last 189 features back to `(3, 3, 21)` before passing that through the Betti MLP. Since the ResNet's output is fixed at 3 features while the Betti compressor's output dimension is a gridsearched hyperparameter, the resulting summary vectors of both those embeddings are unequal in dimension. They are concatenated and then passed through a final fully connected network, which looks like Betti's embedding network (linear, layer normalization, Leaky ReLU, dropout) and it compresses the combined summary statistics down to the target summary dimension used by the normalizing flow. This way, each data type is processed by its own specialized embedder to extract as much information as we can, instead of concatenating raw observations (the full 489) and relying on a generic network to learn both structures at once.

5.3.8 SBI Ensembles

A single-model posterior is sensitive to the random weight initialization and its train/validation split seen during training. Varying the random number generator with similar training gives a posterior that can have better or worse calibration even if both share the same architectures and hyperparameters. To mitigate this, we select the 5 best calibrated ones, out of the 30 trained trials for a given data type. These are combined using `sbi`'s EnsemblePosterior object, which averages the probability densities of the selected posteriors when they're sampled. We get a mixture distribution that can generally discard any artefacts that could occur for a single-model's posterior.

Ensembling is performed independently for each of the three data types (PS2D, Betti and joint PS2D+Betti), rather than mixing posteriors trained on different data types. We do not train one data type, then another, and then mix their posteriors, since they're each trained on their own specific observations and could be uncorrelated. Averaging the posteriors would not give a mixture distribution that is informative of a jointly trained model, it would only show two independent posteriors with their own idea about θ.

Ensembles in SBI smooth the distribution of the posterior and as such there are two ways to think about them in regards to the desired results. On one hand, a wider less constrained posterior could be preferable to an overconfident single-model posterior, especially since our models are trained on simulated data; a posterior that is wider may generalize the real SKA observations better in this case. On the other hand, it is computationally more expensive to store and sample 5 separate posteriors rather than a single one, which may not be justified if a single model is already well calibrated enough. The choice is then based on the speed of sampling, on how constrained we want the posteriors to be and the calibration of the model.

One complication with NPE-based models (but practically only seen when ensembling posteriors), is rejection sampling from the prior (as discussed in the previous Chapter). A mixture distribution could be placing a density mass near the prior's boundaries (see `sbi` documentation's FAQ on this issue Tejero-Cantero et al. 2026). To mitigate this, we set for the ensemble pipeline the prior into the full range between 0 and 1, rather than the dataset-derived one, which is only marginally wider than the empirical θ range (Chapter 3) and so does not meaningfully change the training distribution.

Chapter 6

Results and discussion

6.1 SBI PS2D

6.1.1 Hyperparameters

After gridsearching, we have found these to be our most performant parameters with the negative log validation loss at -9.12:

- The summary dimension is fixed at 3 as that is our ResNet prediction output dimension, consequently only the next two parameters were gridsearched.
- The number of bins for the density estimator: 32
- The learning rate: 0.0001

Note that the negative log validation loss did not vary much for these parameters, with the minimum being at -8.46. The normalizing flow seems to be insensitive to the tested hyperparameter combination for PS2D, meaning that the embedding is the dominant factor as the validation loss is quite good as we will see later for the other models.

6.1.2 30 run trial TARP calibrations

Figure 6.1 shows a consistent S-shaped deviation from the diagonal for all 30 trials. It seems credible regions are overconfident at low credibility levels, and underconfident at high credibility levels. The reoccurence of the same shape for all 30 independently trained models shows it could either be an architectural property or determined by the nature of the observation. The p-values range from 0.596 (trial 18) to 0.979 (trial 20). Our selected model (trial number 20) is thus selected as the best-calibrated model and is used for all evaluations as will follow.

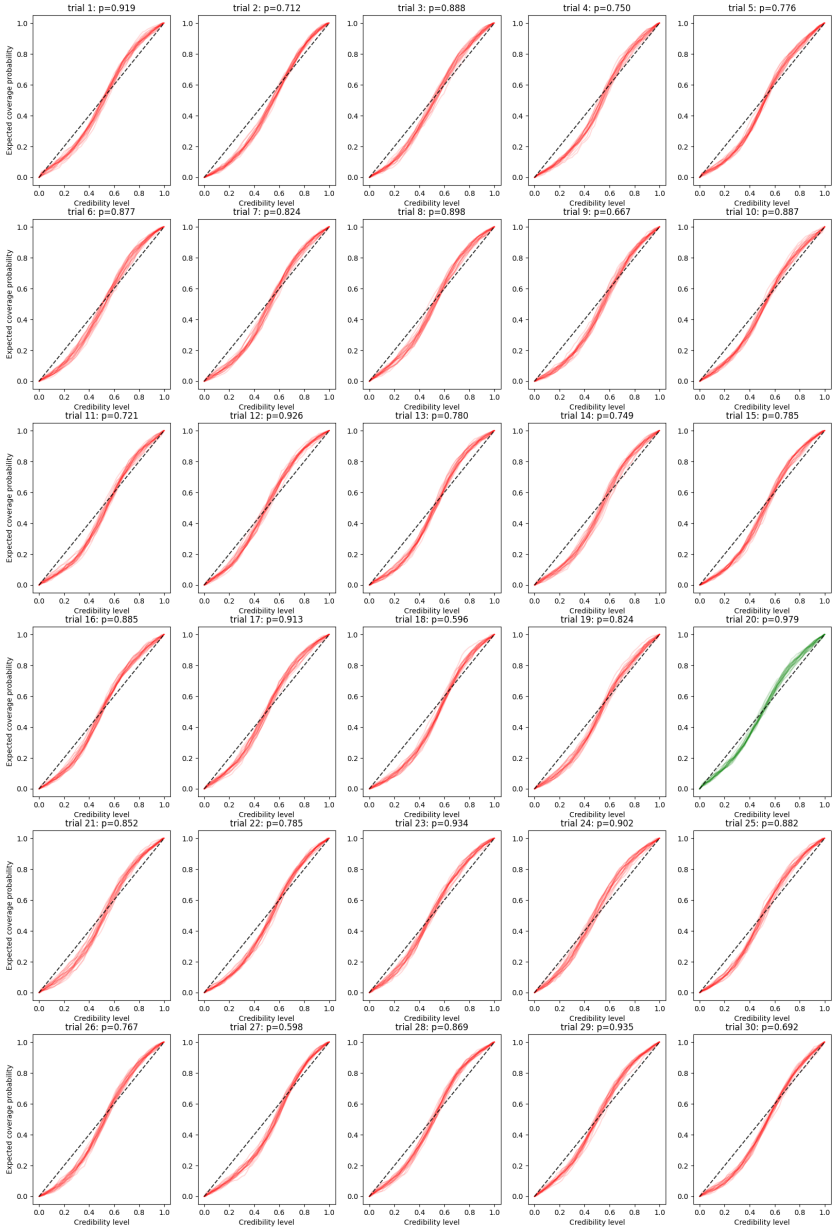
6.1.3 Final TARP calibration

When running TARP again for our best model, on the global test set, we obtain a p-value of 0.951. This is slightly lower than on the validation set which has a p-value of 0.979. If we take an α of 0.05, as is standard, we can observe that the null hypothesis of the model being well calibrated is not rejected. Figure 6.2 shows us the same S-curve deviation.

6.1.4 Posterior predictive check

Table 6.1 and Figure 6.3 shows us a difference in performance for the three reionization scenarios. The low and mid scenarios have a fractional residual close to 0.10, indicating that it is close to the ground truth observation. The high (slow reionization) scenario is worse, with a fractional residual of 0.485 and a mean z-score of 0.994. Its nearest-neighbor distance (0.006), however, is small, comparable to the mid scenario. This could mean the points in parameter space could fetch other close points, but the predicted proxy observations could not match the true ground truth observations.

Figure 6.1: PS2D 30 run trial TARP calibrations



Visually speaking, Figure 6.3's high scenario panel shows that most points are centered near the origin while the true observation can be seen far from the diagonal line, indicating underprediction (the predicted values are much smaller than the true observations). We can refer back to our table and figure in Chapter 3 (Table 3.1 and Figure 3.1) where the slow reionization had low amounts of samples compared to the other two scenarios. The posterior, as we will see, is narrow but biased, and the predicted observation has problems mapping back accurately consistently.

Figure 6.2: PS2D TARP on the global test set

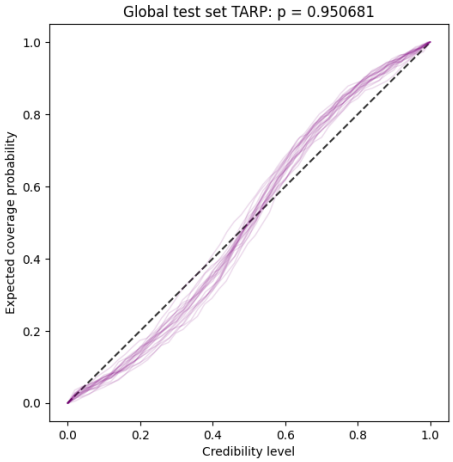
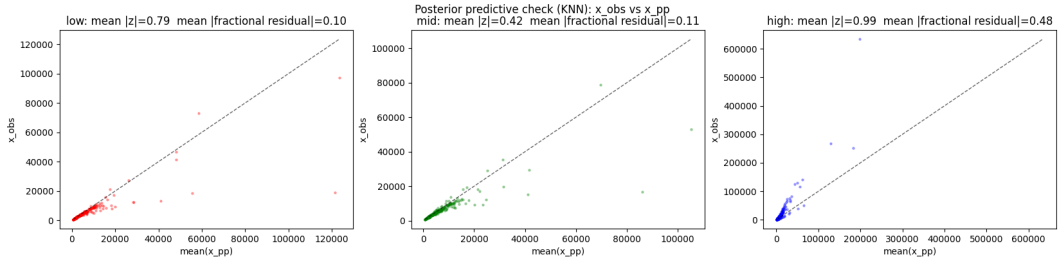


Table 6.1: PS2D PPC metrics

Metric	Low	Mid	High
Mean z	0.789	0.418	0.994
Fractional residual	0.099	0.108	0.485
nearest neighbor distance	0.012	0.006	0.006

Figure 6.3: PS2D PPC for low, middle and high scenarios respectively



6.1.5 Reionization history

Figure 6.4 confirms that the inferred reionization history follows a downward trend for all three scenarios, meaning that the x_{HI} gets ionized as the EoR progresses. The low and mid scenarios show an overestimation of the posterior mean (relative to the ground value) at every frequency, although the true value remains in the credible interval of one standard deviation of the inferred posteriors, except for the mid posterior at the first frequency bin. The high scenario, in this case, shows the closest alignment relative to the ground truth, although we saw with the PPC metrics it has trouble mapping back, so it may be biased.

6.1.6 Posteriors per scenario

Figure 6.5 shows more constrained and taller posterior densities for the high scenario than for low or mid. This could indicate that the model is more confident when estimating the parameter for this specific scenario as we can see in the reionization history plot. One thing we can note is that the PPC results for the same observation showed that this scenario had the largest discrepancy compared to the two others. This means the model's parameter estimation for this scenario is accurate and constrained (confident), yet the predicted observable does not map back consistently.

Figure 6.4: PS2D reionization history

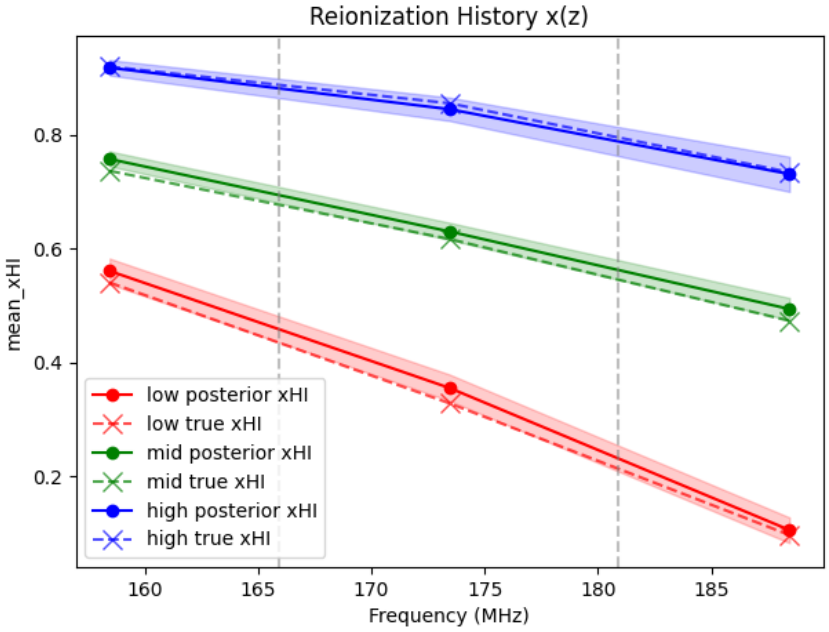
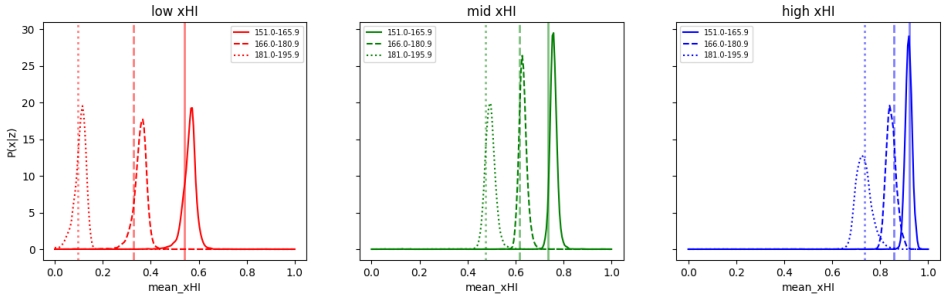


Figure 6.5: PS2D Posteriors per scenario



6.1.7 Pairplots per scenario

Figure 6.6 shows positive correlations for all redshift bins with each other, as seen in the joint distributions in the 3 top right panels in each scenario panel. This means the model has implicitly learned that x_{HI} evolves as the EoR progresses across the redshift bins. It is implicit as we have not explicitly imposed this information within our architecture. In each panel, the true parameter value (in red) is located within, or very near, the dense region of the joint posterior. This also confirms that the global TARP calibration result is indeed good.

6.1.8 Metrics

After evaluation, we retrieve the scoring from Table 6.2. While the R^2 score increases per redshift bin from z_1 to z_3 (98.67% to 99.52%), we can refer back to Chapter 3's θ analysis where z_3 had a larger variance. Since R^2 is normalized by this variance, part of the improvement can be explained by the spread of θ compared to earlier redshift bins, and less so by having a reduction in errors. We can also observe that RMSE and MAE have a peak at z_2 and not at z_1, which could indicate that the inference error does not follow the same trend as R^2 indicates. The ratio between RMSE and MAE is relatively similar across the three bins (1.48-1.60), and this means there's not a redshift bin with disproportionately poor predictions and that they have comparable degrees of error affected by outliers.

Figure 6.6: PS2D pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization

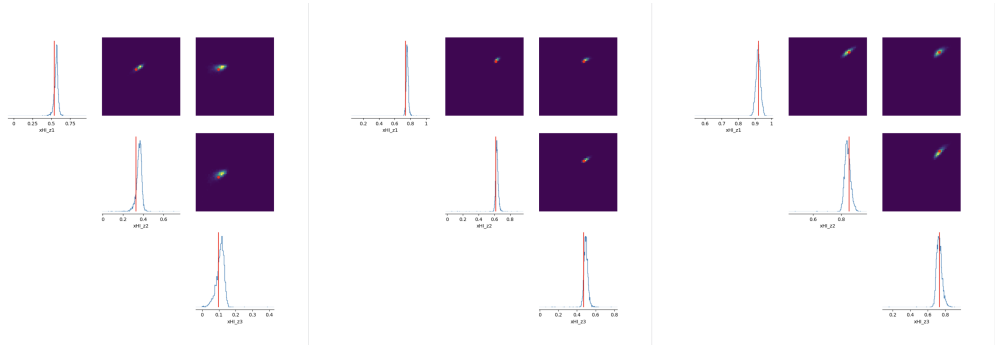


Table 6.2: PS2D scoring metrics

Metric	z_1	z_2	z_3	Overall
R^2 [%]	98.67	99.20	99.52	99.13
RMSE [$\cdot 10^{-3}$]	21.37	21.39	19.84	20.87
MAE [$\cdot 10^{-3}$]	13.33	14.42	13.16	13.64

6.2 SBI Betti

6.2.1 Hyperparameters

After gridsearching, we have found these to be our most performant parameters with a mean validation loss of -14.79:

- The summary dimension for the embedding MLP at 32
- The number of bins for the density estimator at 32
- And the learning rate at 0.0005

We found that the learning rate provided the biggest impact on the performance of the model, with the summary dimension and number of bins providing a marginal difference.

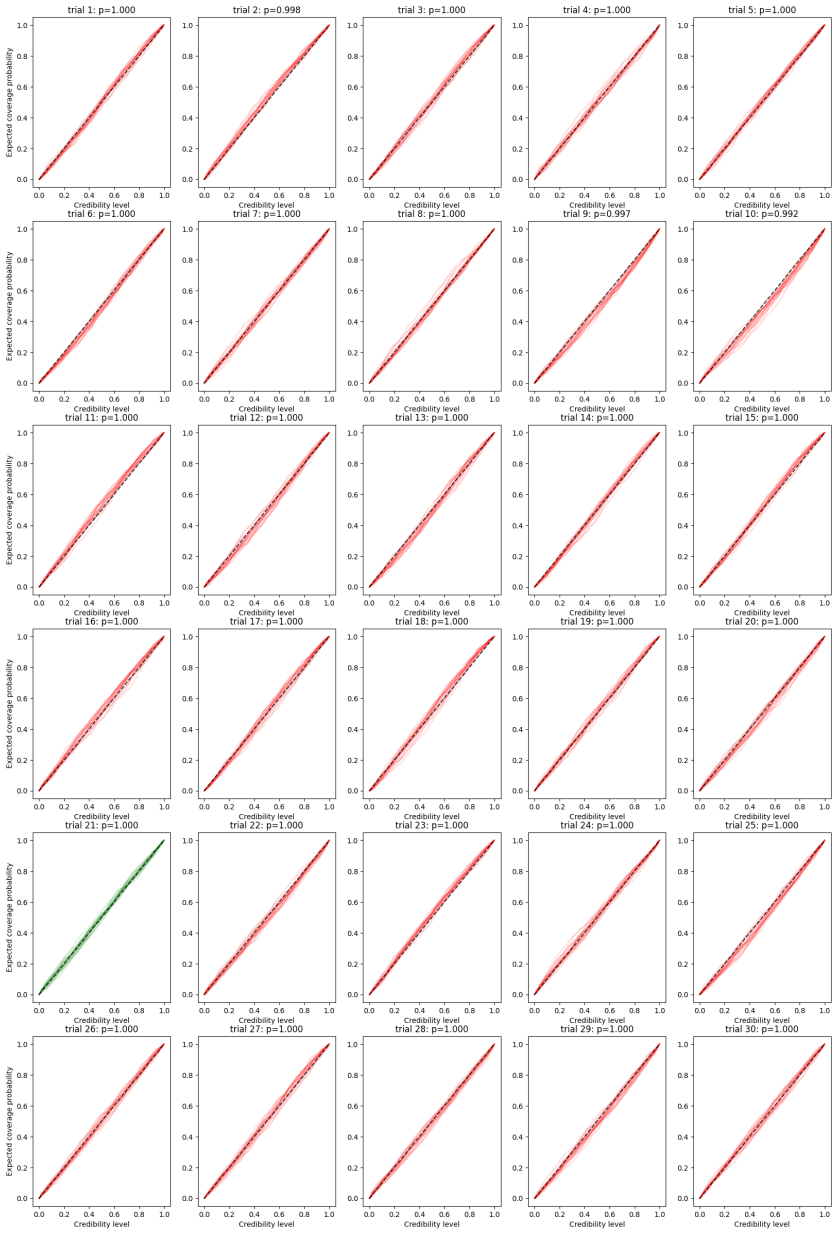
6.2.2 30 run trial TARP calibrations

By referring to Figure 6.7, the plot shows almost perfect calibration for all 30 trials. The p-values range from 0.992 to 1.000. Our selected one (trial number 21) possesses a p-value close to 1.000, and is thus selected to be our best model. They all closely align to the diagonal and have no apparent systematic bias, such as the previously seen S-shaped miscalibration for the PS2D trials. This could suggest that the Betti embedding has better summary capabilities than the ResNet has for the PS2D observations (perhaps aided by the larger output dimension), but this could also be thanks to the larger batch size, as we have previously had worse calibrations (similar in range as PS2D) for smaller batch sizes for Betti observations.

6.2.3 Final TARP calibration

When running TARP again for our best model, on the global test set, we obtain a p-value of 99.99%. The null hypothesis is therefore not rejected. Figure 6.8 shows us curves mostly matching the diagonal. Once again the validation TARP shows a marginally better calibration than on the global test set, though it is negligible. This means the model is well calibrated and reliable for unseen observations.

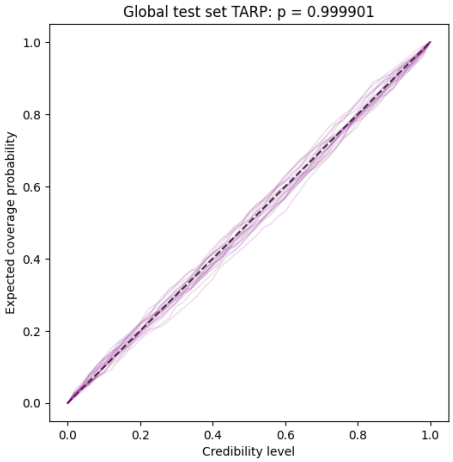
Figure 6.7: Betti 30 run trial TARP calibrations



6.2.4 Posterior predictive check

The TARP shows better calibration than for PS2D, however, the PPC metrics in Table 6.3 are worse across all three scenarios. For example, the low scenario's fractional residual is about three times larger for Betti (0.294) than for PS2D (0.099). The nearest neighbor distance remains the same (quite small) so we can rule out KNN as the source of this difference. The high scenario has comparable metrics which shows both of those models struggle fitting the slow reionization samples. We can also see in the Figure 6.9 that the predicted proxy observations matches well with the low and mid scenarios, but

Figure 6.8: Betti TARP on the global test set

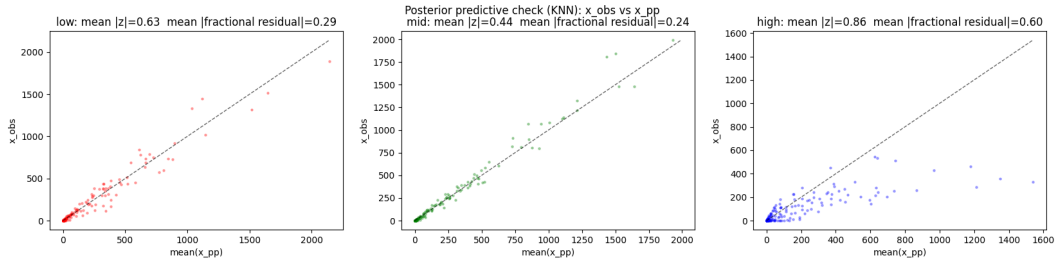


this time predict values that are too high compared to the true observations for the high scenario.

Table 6.3: Betti PPC metrics

Metric	Low	Mid	High
Mean z	0.631	0.436	0.860
Fractional residual	0.294	0.236	0.597
nearest neighbor distance	0.012	0.007	0.007

Figure 6.9: Betti PPC for low, middle and high scenarios respectively



6.2.5 Reionization history

Figure 6.10 shows the same downward trend, but the low and mid are this time closely aligned to the ground truth, and very constrained, with a little overestimation at z_1 for the low scenario. The high scenario is different than the one observed for the PS2D posterior: while the latter had a near perfect alignment, the Betti posterior mean here underestimates the true value more as the EoR progresses (the gap widens).

6.2.6 Posteriors per scenario

Figure 6.11 shows that low and mid scenarios have the posteriors with the most constrained and confident posteriors (with low scenario's at z_3 being almost a pinpoint prediction). The high scenario is wider and could be explained by the PPC's overprediction of observables for this specific scenario.

Figure 6.10: Betti reionization history

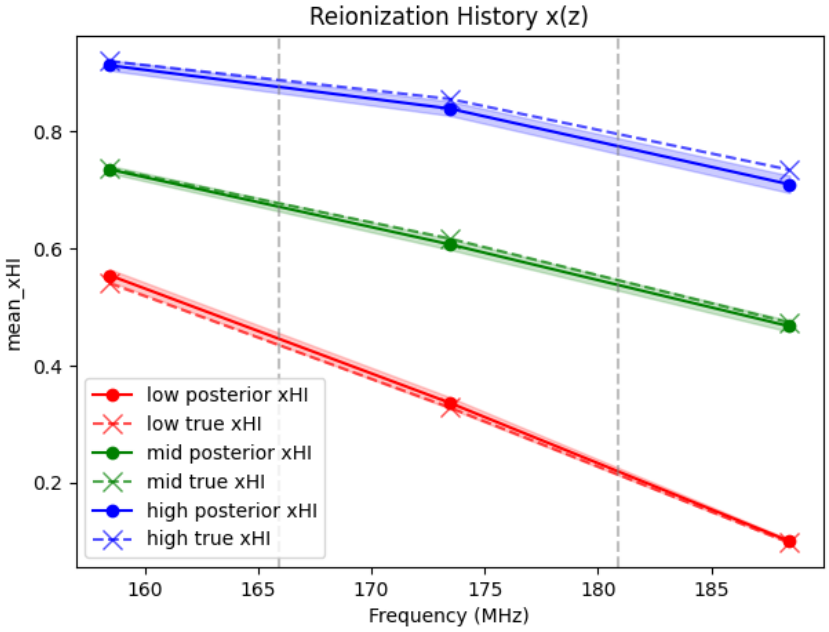
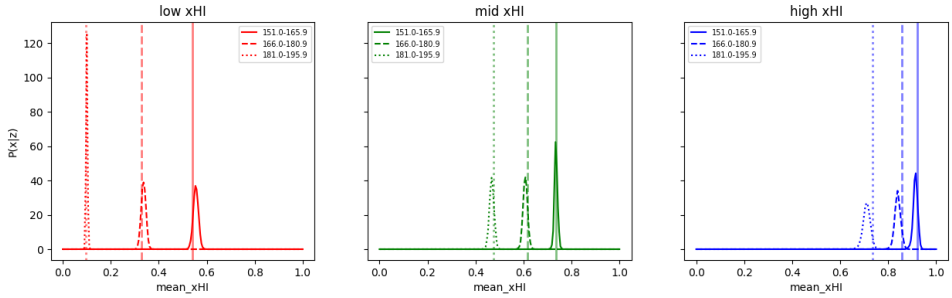


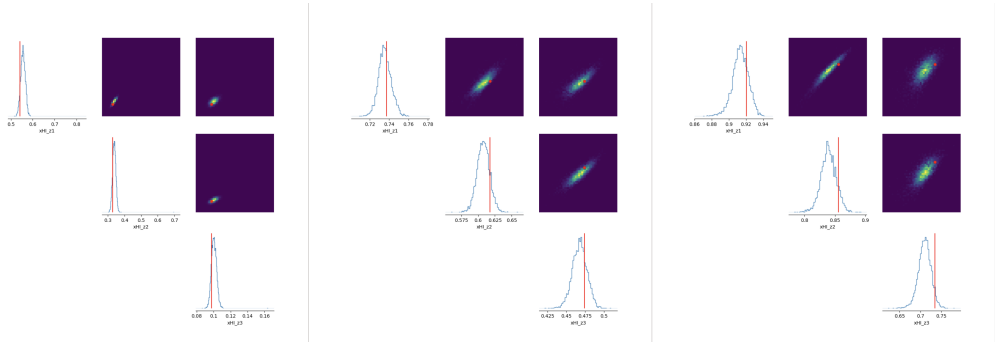
Figure 6.11: Betti Posteriors per scenario



6.2.7 Pairplots per scenario

Figure 6.12's joint plots (especially for the mid and high scenarios) show strong correlations between posteriors of redshift bins. They look wider and more elongated than the ones shown for PS2D, but that's the automatic zoom level from sbi's pairplots function.

Figure 6.12: Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization



6.2.8 Metrics

After evaluation, we retrieve the scoring from Table 6.4 and observe that the R^2 score is more uniform across redshift bins (from 99.15% to 99.57%) than for PS2D which is more spread out. Inference quality is fairly consistent for this data type. RMSE increases from z_1 to z_3 (17.06 to 18.83) while MAE has a peak at z_2 similar to PS2D and dips again for z_3. Since RMSE is more sensitive to large individual errors than MAE, the divergence could be explained by the larger amount of outliers at the latest stage of reionization. The ratio of RMSE/MAE goes from 1.54 to 1.76 as the EoR progresses so this could also explain the outlier phenomenon at this redshift bin of z_3.

Table 6.4: Betti scoring metrics

Metric	z_1	z_2	z_3	Overall
R^2 [%]	99.15	99.42	99.57	99.38
RMSE [$\cdot 10^{-3}$]	17.06	18.32	18.83	18.07
MAE [$\cdot 10^{-3}$]	11.09	11.65	10.67	11.14

6.3 SBI PS2D + Betti

6.3.1 Hyperparameters

After gridsearching, the best performing hyperparameters were:

- A summary dimension of 16
- The number of bins at 32
- and the learning rate at 0.0005

The summary dimension is lower than for Betti (which was at 32), while the other hyperparameters and the resulting validation loss were comparable across both.

6.3.2 30 run trial TARP calibrations

Figure 6.13 shows us softer S-curve deviations which we saw for all PS2D trials. The p-value ranges from 0.946 to 1.000 which is about the middle-of-the-road between PS2D (0.596-0.979) and Betti (0.992-1.000), being closer to the latter. This means that parameter-space calibration for PS2D improves when coupled with Betti numbers relative to PS2D alone, as found in Cerardi et al. 2025's finding that combining the power spectrum with higher-order statistics improves the constraining power (see Chapter 2). Trial 6 at p=1.000 is selected as the best-calibrated model.

6.3.3 Final TARP calibration

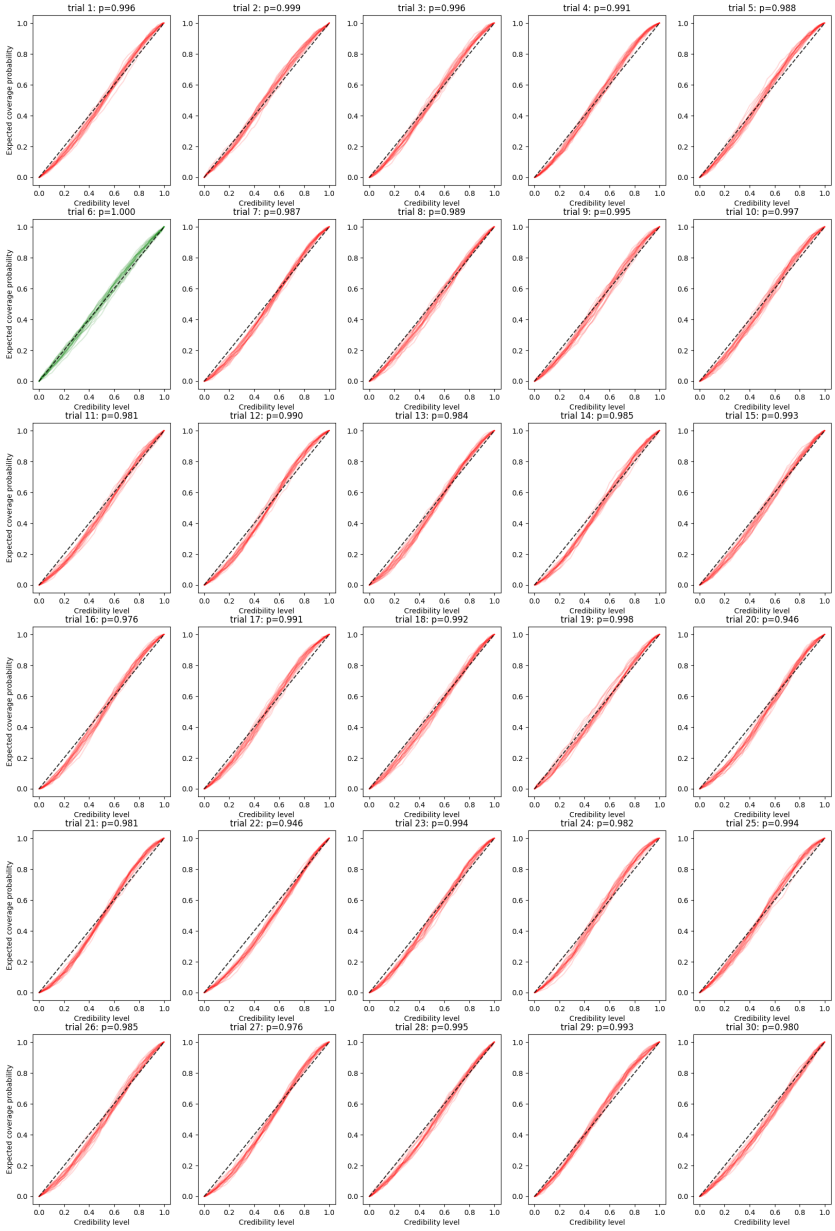
Figure 6.14 shows the global TARP at 99.81%, confirming a strong calibration which is comparable to Betti's 99.99% and much stronger than PS2D's 95.07%.

6.3.4 Posterior predictive check

Table 6.5 shows a couple of interesting observations:

1. For the low and mid scenarios, the joint model's fractional residuals (0.144, 0.156) are located between PS2D's (0.099, 0.108) and Betti's (0.294, 0.236), meaning the reliability in observable-space of pairing the two data types is intermediate for those two scenarios.
2. The high scenario is however not intermediate. The fractional residual of 0.822 and mean z-score of 1.498 exceeds that of PS2D (0.485, 0.994) and Betti (0.597, 0.860). Of the three trained posteriors, it has the most trouble inferring back to a predicted observable that matches reality.

Figure 6.13: PS2D + Betti 30 run trial TARP calibrations



The nearest-neighbor distance at 0.005 is consistent with the other data types, so that cannot be the root cause of this excess.

This result means that the joint model has a parameter-space calibration close to Betti's, but the observable-space exceeds both models greatly.

Figure 6.14: PS2D + Betti TARP on the global test set

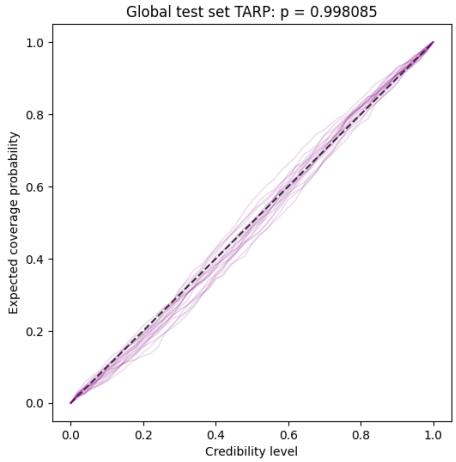
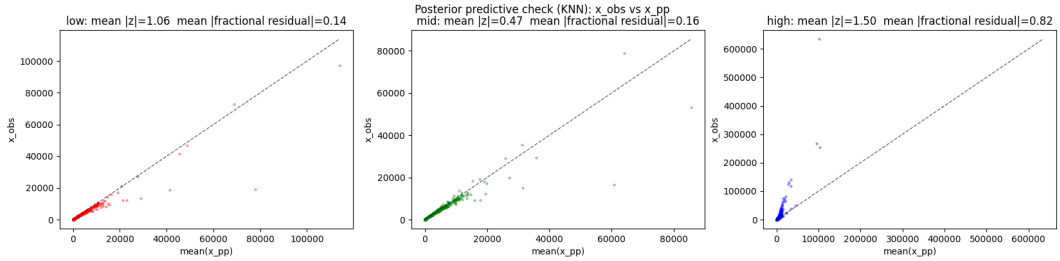


Table 6.5: PS2D + Betti PPC metrics

Metric	Low	Mid	High
Mean z	1.063	0.473	1.498
Fractional residual	0.144	0.156	0.822
nearest neighbor distance	0.013	0.006	0.005

Figure 6.15: PS2D PPC for low, middle and high scenarios respectively



6.3.5 Reionization history

As with Betti (and unlike PS2D), the joint model's posterior mean shown in Figure 6.16 underestimates the high scenario's true value. Given that the TARP calibration also closely resembles Betti's, it suggests that the joint model has learned this same behaviour from the Betti component rather than averaging it with PS2D's contribution, although we do see that it does not diverge from z_1 to z_3 as with Betti. This could also potentially be explained by the summary dimension of Betti's at 16 being more predominant when passed to the fully connected final layer than PS2D's 3-dimensional summary statistic.

6.3.6 Posteriors per scenario

The given posteriors shown in Figure 6.17 very closely match those seen for Betti although there is a subtlety: the y-axis indicates the confidence of those posteriors and those of Betti's reached high, especially for the low scenario. The ones for the joint model have a smaller confidence, reaching to 80 for the highest peak seen for the low scenario, whereas the same one for Betti reaches above 120.

Figure 6.16: PS2D + Betti reionization history

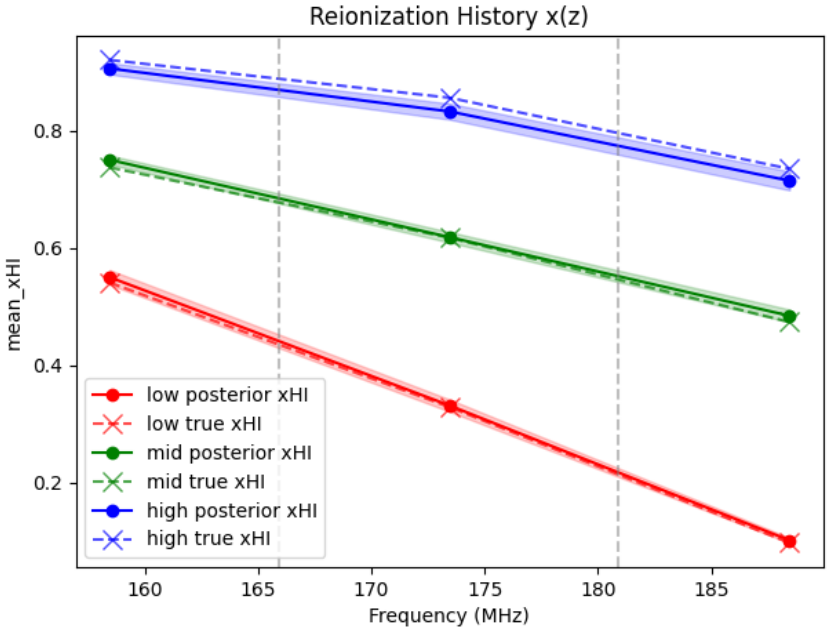
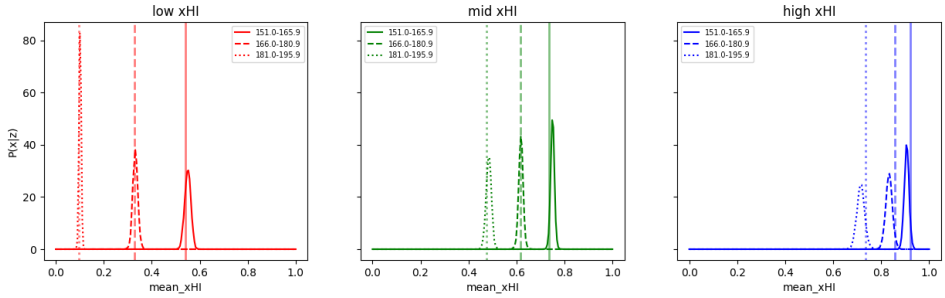


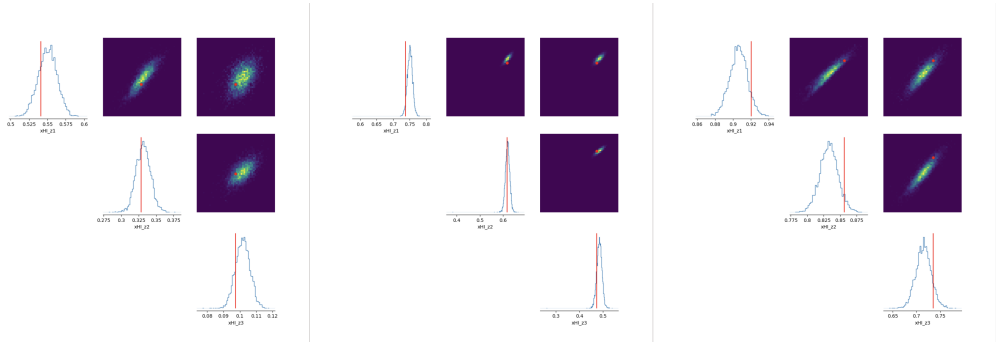
Figure 6.17: PS2D + Betti Posteriors per scenario



6.3.7 Pairplots per scenario

Figure 6.18 shows diagonal elongated regions for the joint distributions for the low, mid and high scenarios, which matches the one seen for Betti and indicates good correlation between the bins.

Figure 6.18: PS2D + Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization



6.3.8 Metrics

After evaluation, we retrieve the scoring from Table 6.6. Unlike PS2D and Betti individually, where R^2 and the absolute error metrics showed that they were diverging from each other across redshift bins, the joint model shows all three metrics improving step after step from z_1 to z_3 (R^2 increases, RMSE and MAE decrease) which indicates a more global reduction in error (normal and outlier). Training both data types jointly gives better inference power with less errors.

Table 6.6: PS2D + Betti scoring metrics

Metric	z_1	z_2	z_3	Overall
R^2 [%]	99.15	99.54	99.68	99.46
RMSE [$\cdot 10^{-3}$]	17.06	16.26	16.07	16.46
MAE [$\cdot 10^{-3}$]	10.92	10.80	10.01	10.58

6.4 SBI 5-Ensemble PS2D

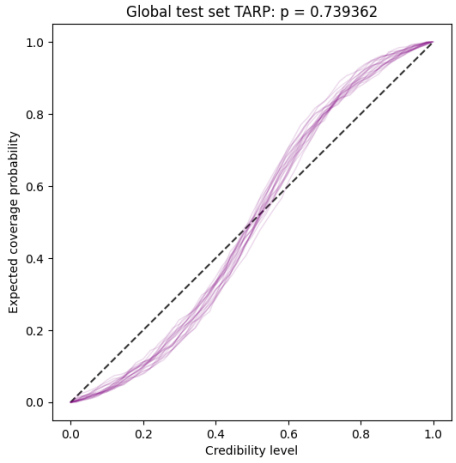
6.4.1 30 run trial TARP calibrations

Figure D.1 in Appendix D shows the same S-shaped deviation observed for all single-model PS2D trials, again present in all ensemble trials. The five best-calibrated trials (7, 22, 23, 25, 30) are selected to form the ensemble.

6.4.2 Final TARP calibration

The ensemble's global test set p-value of 73.94% is much lower than the single best PS2D model's p-value of 95.07% and Figure 6.19 shows a very pronounced S-curve. It seems ensembling when the posteriors are correlated but miscalibrated in the same manner exacerbates miscalibration. Formally, with the standard $\alpha = 0.05$ threshold, it is still well-calibrated, but much less so than the single-model result.

Figure 6.19: Ensemble PS2D TARP on the global test set



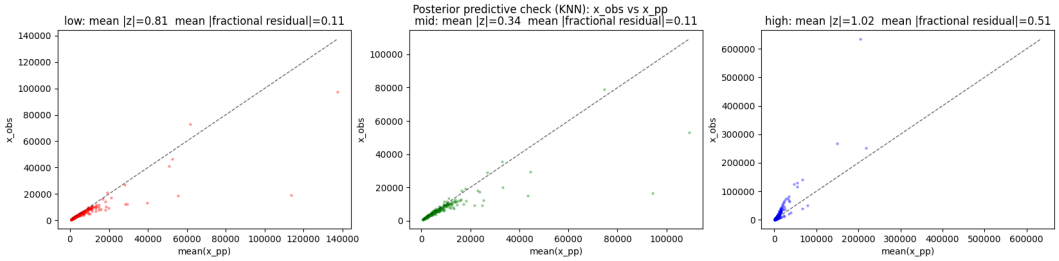
6.4.3 Posterior predictive check

Despite the worsened parameter-space calibration, the PPC metrics in Table 6.7 are almost identical, with the fractional residuals within 0.01-0.03 of one another across all three scenarios. The ensemble's degraded parameter-space calibration does not indicate the same degradation for the observable-space, as it stays consistent.

Table 6.7: Ensemble PS2D PPC metrics

Metric	Low	Mid	High
Mean z	0.809	0.338	1.020
Fractional residual	0.109	0.110	0.511
nearest neighbor distance	0.012	0.006	0.007

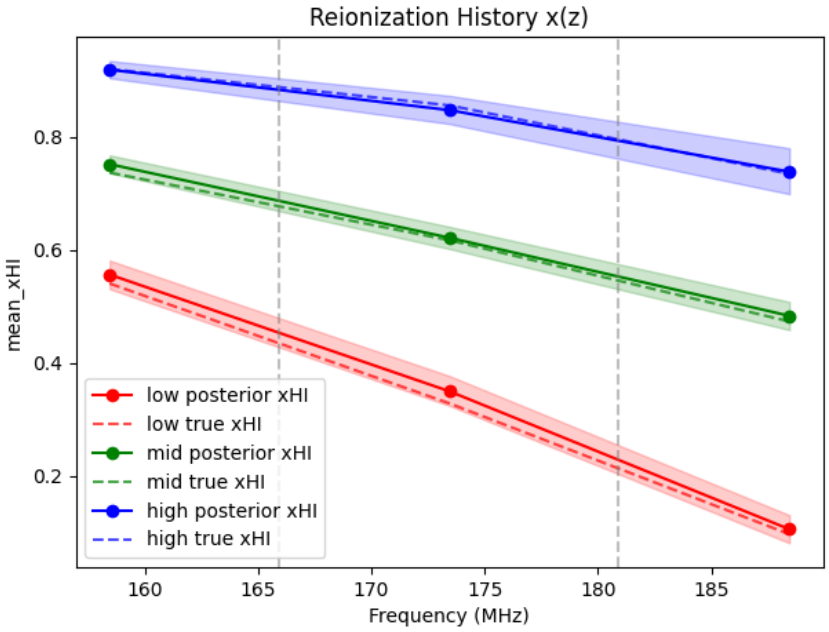
Figure 6.20: Ensemble PS2D PPC for low, middle and high scenarios respectively



6.4.4 Reionization history

Figure 6.21 shows the same pattern as the single-model result. We see a small overestimation for the low and mid scenarios and a close alignment for the high scenario. At z_3, the intervals are wider than in the single-model case.

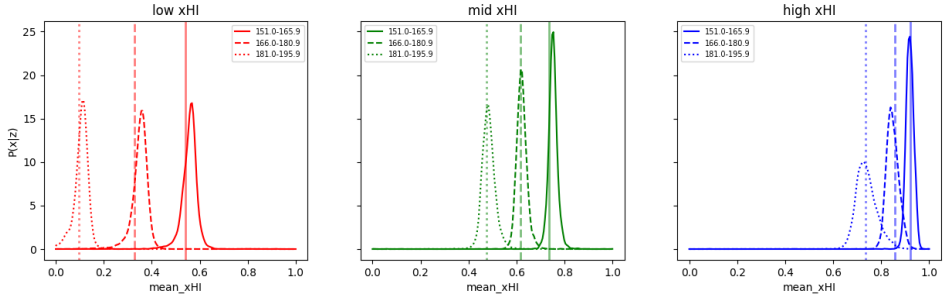
Figure 6.21: Ensemble PS2D reionization history



6.4.5 Posteriors per scenario

The posteriors are nearly identical to the single-model, save for the confidence being a bit lower for all posteriors, as can be seen in Figure 6.22.

Figure 6.22: Ensemble PS2D Posteriors per scenario



6.4.6 Pairplots per scenario

The pairplots in Figure 6.23 shows the same correlated regions observed for the single-model PS2D pairplots, with the true parameter located near or in the high-density regions in all three scenarios.

Figure 6.23: Ensemble PS2D pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization

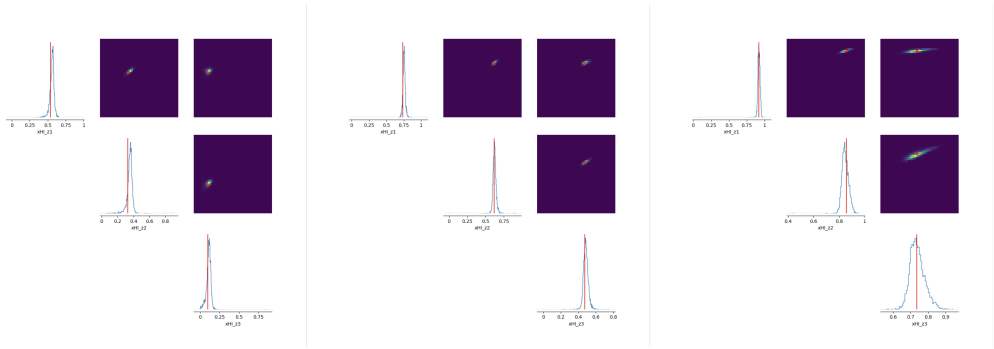
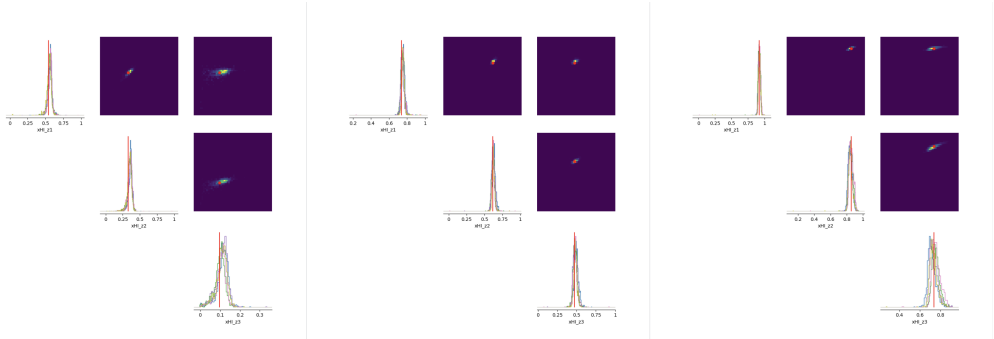


Figure 6.24 shows the five individual posteriors of the ensemble overlaid for each scenario, and we can see that all 5 are largely similar. For z_1 and z_2, they very closely match, whereas there is some slight discrepancy at z_3, especially for the high scenario, which corroborates what we found with the PPC metrics that the high scenario is the least reliable of the three. The observable-space calibration is the weakest there, and we see that the posteriors are less in agreement than with the other scenarios.

Figure 6.24: Ensemble PS2D pairplots of all 5 posteriors per scenario; left: fast reionization, middle: normal reionization, right: slow reionization



6.4.7 Metrics

Table 6.8 shows performance almost identical to the single-model PS2D, with R^2 scores increasing for z_1 with marginal improvements for z_2 and z_3. It seems the earlier epoch benefits more from

ensembling. RMSE and MAE are similar, with the same peak at z_2.

Table 6.8: Ensemble PS2D scoring metrics

Metric	z_1	z_2	z_3	Overall
R^2 [%]	98.78	99.22	99.53	99.18
RMSE [$\cdot 10^{-3}$]	20.40	21.17	19.72	20.43
MAE [$\cdot 10^{-3}$]	13.18	14.69	13.67	13.85

6.5 SBI 5-Ensemble Betti

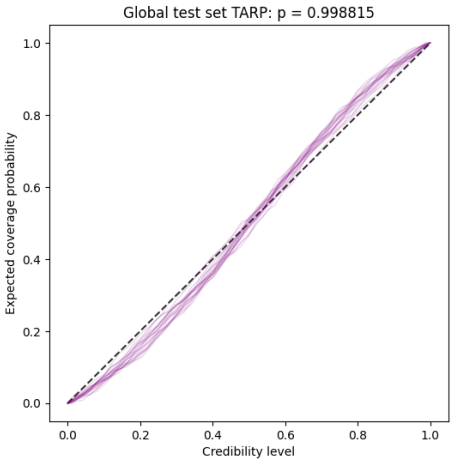
6.5.1 30 run trial TARP calibrations

As we can see on Figure D.2 in Appendix D, the trials show the same near-perfect calibration seen for the single-model Betti trials, with p-values going from 0.996 to 1.000. The five best calibrated trials are 4, 7, 8, 22, and 23 were selected to form the ensemble.

6.5.2 Final TARP calibration

PS2D showed a substantial drop when ensembling for the global TARP p-value, whereas the Betti ensemble's global p-value of 99.88% is only marginally worse than the single-model's p-value of 99.99%, as seen in Figure 6.25. An interesting observation is that it produces the same S-curve (albeit very slight), meaning it is more of an architectural behaviour, rather than based on PS2D's observations. The same mechanism is in play, where averaging that comes from ensembling the posteriors comes with worsening the miscalibrated areas, although in this case it is a very slight miscalibration.

Figure 6.25: Ensemble Betti TARP on the global test set



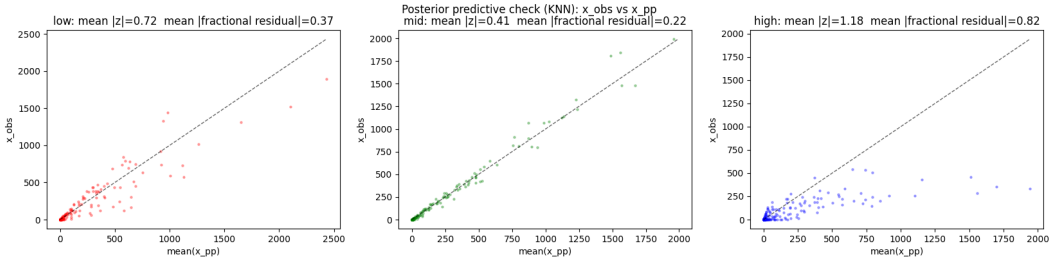
6.5.3 Posterior predictive check

Table 6.9 shows more degradation in observable-space compared to the single-model Betti result, for both low (fractional residual from 0.294 to 0.366) and high (0.597 to 0.823) scenarios. The mid scenario remains stable. As with the other models, the nearest neighbor distance remains small. We see a bit of the reverse of what we saw for the PS2D ensemble: the TARP calibration remains good, while the PPC metrics are worsened, whereas for PS2D the TARP calibration worsened, while the PPC metrics remained stable (comparatively). It seems ensembling worsens any substantial artefacts present in both parameter and observable-space.

Table 6.9: Ensemble Betti PPC metrics

Metric	Low	Mid	High
Mean z	0.721	0.406	1.183
Fractional residual	0.366	0.224	0.823
nearest neighbor distance	0.012	0.007	0.006

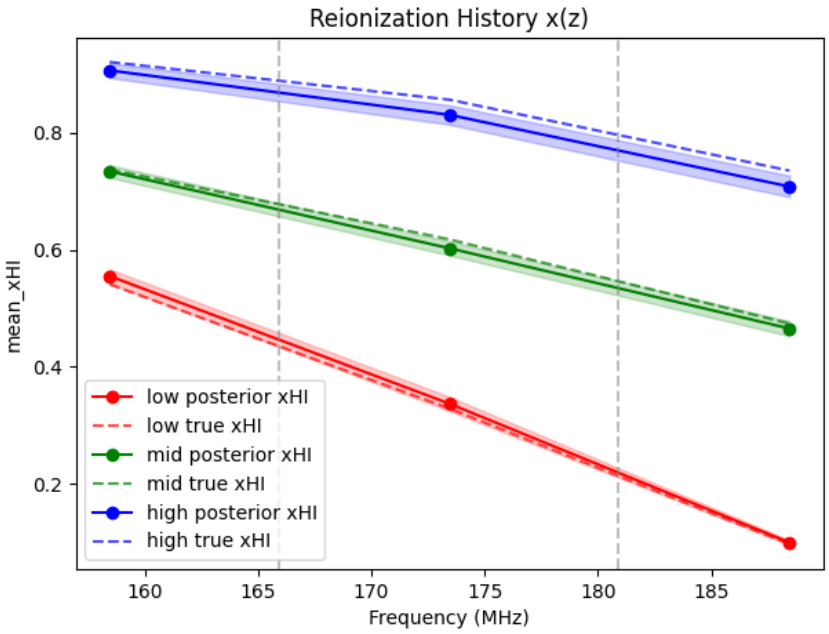
Figure 6.26: Ensemble Betti PPC for low, middle and high scenarios respectively



6.5.4 Reionization history

Figure 6.27 shows the same pattern observed for the single-model Betti, where the high scenario is underestimated, although this time it seems it does not diverge as much as the EoR progresses. The mid scenario's z_2 also seems slightly underestimated compared to the ground truth.

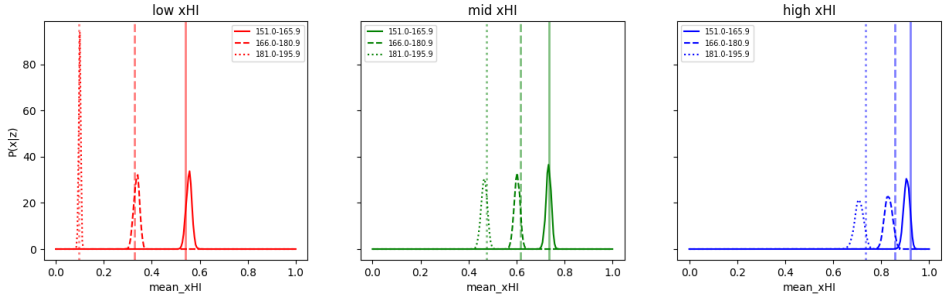
Figure 6.27: Ensemble Betti reionization history



6.5.5 Posteriors per scenario

Figure 6.28 shows us the same behaviour as for the PS2D ensemble, where the confidence in the posterior is a bit lower. We also see that, comparatively, the posteriors are closer with their peaks than for the single-model Betti, except for the low scenario's z_3 posterior.

Figure 6.28: Ensemble Betti Posteriors per scenario



6.5.6 Pairplots per scenario

Figure 6.29 shows the same good correlation as before. One thing to note is that for the mid scenario's correlation between z_1 and z_2, it seems that the ground truth point falls just outside the density region, which could explain the underestimation at z_2 seen in the reionization history's plot.

Figure 6.29: Ensemble Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization

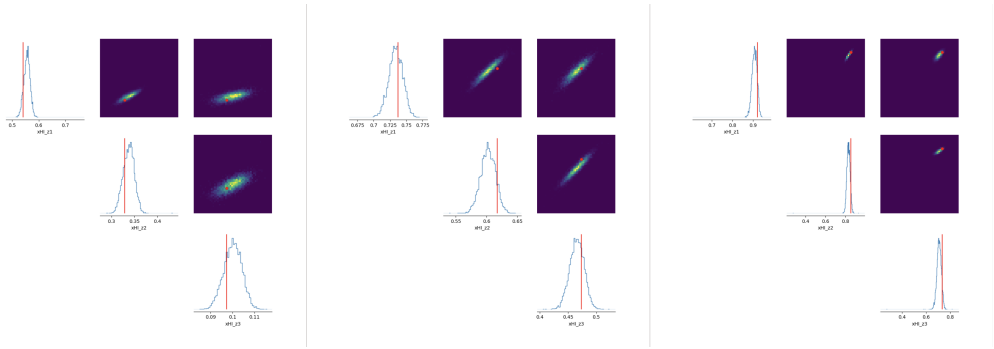
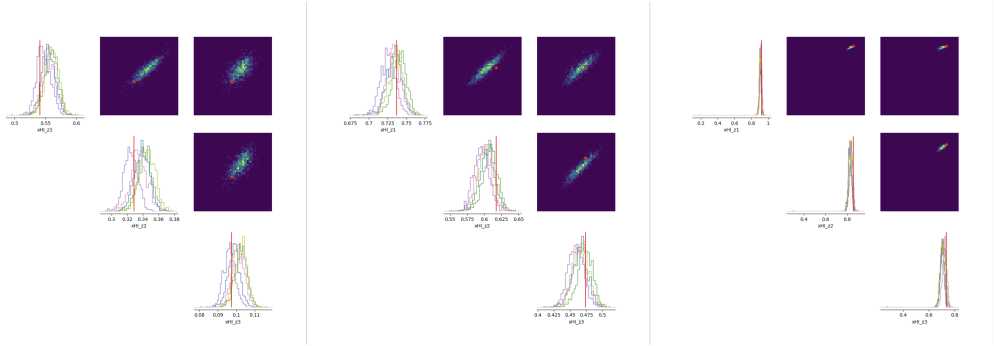


Figure 6.30 shows us the opposite behaviour compared to the posteriors composing the PS2D ensemble: we can see that many posteriors do not agree with one another much, except for the high scenario.

Figure 6.30: Ensemble Betti pairplots of all 5 posteriors per scenario; left: fast reionization, middle: normal reionization, right: slow reionization



6.5.7 Metrics

The scoring from Table 6.10 shows us that R^2 has improvements for each redshift bin compared to the single-model Betti results, as well as RMSE and MAE, especially RMSE at z_3 going from 18.83 for the single-model to 16.78. This could mean that the Betti data type benefits from ensembling more than PS2D.

Table 6.10: Ensemble Betti scoring metrics

Metric	z_1	z_2	z_3	Overall
R^2 [%]	99.19	99.49	99.66	99.45
RMSE [$\cdot 10^{-3}$]	16.66	17.10	16.78	16.85
MAE [$\cdot 10^{-3}$]	10.63	10.84	9.92	10.46

6.6 SBI 5-Ensemble PS2D + Betti

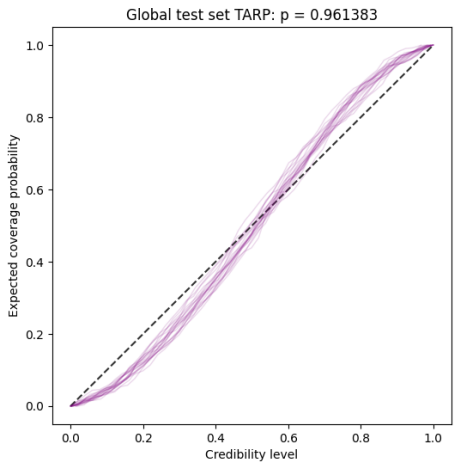
6.6.1 30 run trial TARP calibrations

Figure D.3 in Appendix D shows p-values from 0.925 to 1.000. Five models were selected (7, 13, 26, 27 and 30). The 30 trials also show an S-curve, although milder than seen for PS2D and its ensemble and comparable to the joint single-model.

6.6.2 Final TARP calibration

Figure 6.31 shows a degradation of the global test set p-value of 96.14% compared to the single-model joint p-value of 99.81% as is consistent with the other ensemble models. PS2D shows a strong calibration loss when ensembling, Betti shows a marginal loss and the joint ensemble shows a loss intermediate of those two.

Figure 6.31: Ensemble PS2D + Betti TARP on the global test set



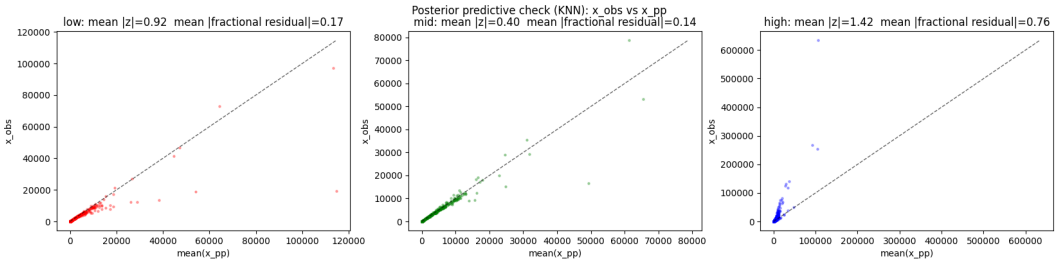
6.6.3 Posterior predictive check

With the metrics presented in Table 6.11, they show results close to the joint single-model, with a small improvement for mid and high, and a small degradation for low's fractional residual (0.144 to 0.170). The high scenario remains stable and the fractional residual is slightly better than Betti's ensemble (0.762 for the joint ensemble, 0823 for Betti).

Table 6.11: Ensemble PS2D + Betti PPC metrics

Metric	Low	Mid	High
Mean z	0.917	0.399	1.419
Fractional residual	0.170	0.139	0.762
nearest neighbor distance	0.013	0.007	0.005

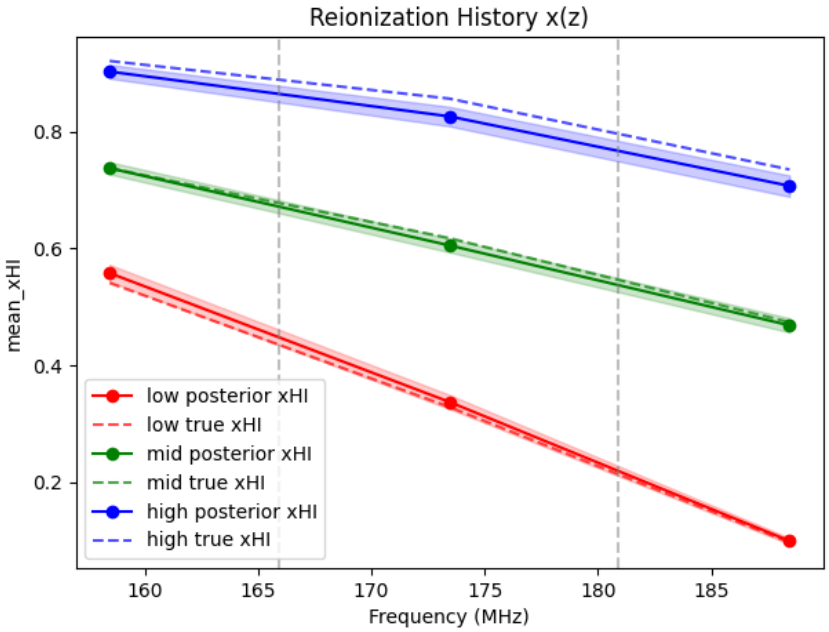
Figure 6.32: Ensemble PS2D + Betti PPC for low, middle and high scenarios respectively



6.6.4 Reionization history

Figure 6.33 shows the same underestimation seen for the joint and Betti single and ensemble models.

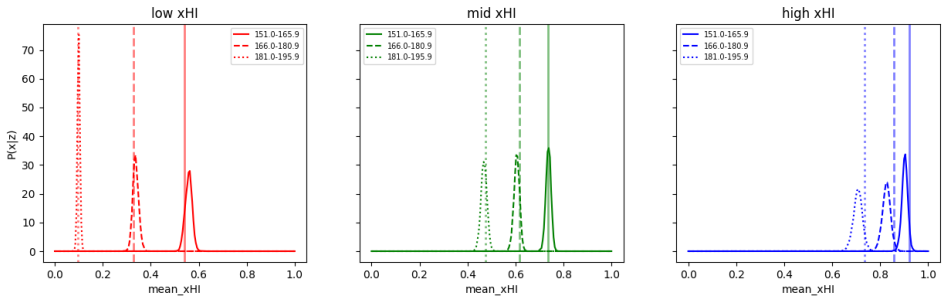
Figure 6.33: Ensemble PS2D + Betti reionization history



6.6.5 Posteriors per scenario

Figure 6.34 shows the same thing as for the other ensembles, with the confidence a bit lower and the peaks more aligned.

Figure 6.34: Ensemble PS2D + Betti Posteriors per scenario



6.6.6 Pairplots per scenario

Figure 6.35 shows good correlations between bins, with the same z_2 artefact as the other ensemble where the ground truth falls very near the density region.

Figure 6.35: Ensemble PS2D + Betti pairplots per scenario; left: fast reionization, middle: normal reionization, right: slow reionization

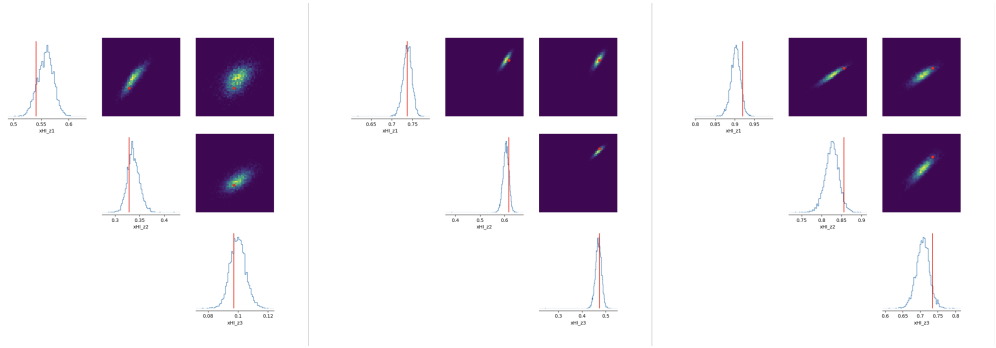
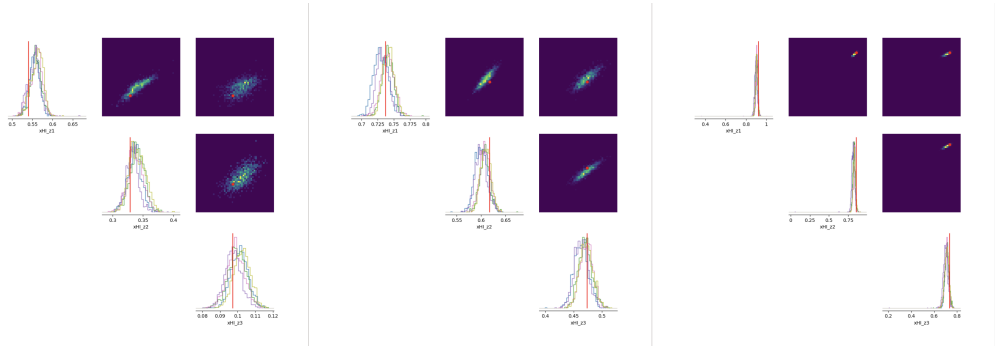


Figure 6.36 shows the same Betti ensemble behaviour where the low and mid have disagreements between posteriors, with the high scenario being very tightly connected. We do see that it is less extreme than with the Betti ensemble which may be explained by the contribution of PS2D.

Figure 6.36: Ensemble PS2D + Betti pairplots of all 5 posteriors per scenario; left: fast reionization, middle: normal reionization, right: slow reionization



6.6.7 Metrics

Table 6.12 shows us the best scores (R^2 99.53%, RMSE 15.41 $\times 10^{-3}$, MAE 9.83 $\times 10^{-3}$), which is an improvement on the single-model joint results.

Table 6.12: Ensemble PS2D + Betti scoring metrics

Metric	z_1	z_2	z_3	Overall
R^2 [%]	99.29	99.58	99.72	99.53
RMSE [$\cdot 10^{-3}$]	15.64	15.49	15.09	15.41
MAE [$\cdot 10^{-3}$]	9.98	10.22	9.23	9.83

6.7 Discussion

Table 6.13: Overview of the model metrics

Model	TARP p-value [%]	z_1 R^2 [%]	z_2 R^2 [%]	z_3 R^2 [%]	Overall R^2 [%]
SBI PS2D	95.07	98.67	99.20	99.52	99.13
SBI Betti	**99.99**	99.15	99.42	99.57	99.38
SBI PS2D + Betti	99.81	99.15	99.54	99.68	99.46
SBI 5-Ensemble PS2D	73.94	98.78	99.22	99.53	99.18
SBI 5-Ensemble Betti	99.88	99.19	99.49	99.66	99.45
SBI 5-Ensemble PS2D + Betti	96.14	**99.29**	**99.58**	**99.72**	**99.53**

The validation-set TARP p-value is always higher than the global test set p-value for all three single-model pipelines. Each selected trial was chosen because it achieved the highest mean TARP p-value among 30 candidates, each itself sampling noise across 20 realizations (Section 4.2.3). Selecting the maximum of 30 noisy estimates tends to favour a trial that got a lucky draw thanks to its particular validation split, and statistically this also means that the spread among the thirty trials in their mean p-value would inform on the calibration loss when evaluated against the untouched global test set. PS2D, whose 30 trials went from p=0.596 to p=0.979, shows the largest gap (from 0.979 to 0.951), while Betti, whose 30 trials are similar and near p=1.000, shows a negligible one (global p-value remains at 0.9999). The joint model, with an intermediate spread, shows an intermediate gap.

We have also found that batch size affects calibration, with the two data types exhibiting opposite behaviours with a batch size of 512, compared to the final size of 4096. It is possible that the statistical variance of PS2D benefits more from training with smaller batches, whereas Betti numbers need more. It could also be an architectural phenomenon, where the flow's own batch size only ever affects a jointly-trained embedding directly, since Betti's compressor receives gradient for that same batch. For PS2D, the ResNet is pretrained separately at a fixed batch size of 512 regardless of the flow's batch size, so this disjointedness could be the cause of the high miscalibration and why the two data types behave differently to the same increase.

Ensembling's viability depends on how much in common the selected trials have in their miscalibrations, since the effects get compounded and the global calibration suffers the heavier the common miscalibrations are. In this report's case, with how we have trained the models, it has practical implications when selecting an ensemble model: ensembling PS2D alone is not recommended, given that each trial shares the S-curve deviation (calibration cost goes from p=0.951 to 0.739) while only getting a marginal improvement for R^2's z_1 score. Betti can ensemble without issue as there is a negligible calibration cost, with some improvement across the redshift bins for the R^2 scores. The joint ensemble offers the strongest scores which may be recommended when inferring the real observations once the SKA comes online as it only has a small calibration cost compared to the joint single-model (p=0.961 compared to 0.998).

In Appendix D, we have the full numerical values for the credible ($\pm 1\sigma$) upper and lower boundaries and the mean neutral reionization fraction x_{HI}. These show proof of how our architectures behave for specific scenarios and redshift bins. For example, the latest stage of reionization in the fast scenario (Low z_3), Betti's credible interval width ($\Delta = \mathrm{Upper} - \mathrm{Lower}$) becomes much smaller, down to 0.0060. This is more than seven times smaller than PS2D, whose interval width is equal to 0.0448. Speaking in cosmological terms, we can expect this difference since, when reionization reaches z_3, the IGM is almost totally ionized ($x_{\mathrm{HI}} \sim 0.10$). While PS2D could be dealing with low signal-to-noise ratios due to not having enough signals being emitted from the almost empty medium, Betti numbers can still count the remaining discrete structures (b_0, b_1 and b_2) without worrying about the noise, and this allows the normalizing flow to constrain the posterior for the latter significantly.

We can also observe that training the two data types jointly creates a structure that gets behaviour from both of them. At that same late stage (Low z_3), Betti's width of 0.0060 is widened slightly in the joint model to 0.0090, still much closer to Betti's own value than to PS2D's wider 0.0448. On the other end, in the slow reionization scenario (High z_3), PS2D's large uncertainty width of 0.0615 becomes half of that in the joint model down to 0.0319, still close to Betti's own 0.0295 in that same

bin. We can validate that the joint embedder does not ignore one data type over the other, where it displays the accuracy at the low scenario of Betti where PS2D is rather weak, and it uses PS2D's variance to slightly widen Betti's constrained posteriors, although it is not a true average of the two.

The slow reionization scenario also shows how the training dataset's distribution affects uncertainty, since the interval width Δ becomes larger as the EoR progresses from z_1 to z_3. This holds for the high scenario for every model (from +44% up to +164%), and also for the mid scenario although more marginally (+16% to +55%). For the single PS2D model, the high scenario width more than doubles from 0.0270 to 0.0615, and for the PS2D ensemble, it jumps by 164% (from 0.0307 to 0.0812). Only the low scenario does not follow this pattern, which stays stable or becomes more constrained at z_3. It seems the uncertainty increases at late epochs regardless of the scenario, and we can lean on our exploratory data analysis in Chapter 3 where most parameters in the training set are $x_{\mathrm{HI}} < 0.70$ at z_3 meaning z_3's distribution is more spread out across the whole dataset than z_1's. Because the normalizing flow only has sparse amounts of parameters to deal with in the higher part of that region, it widens the credible boundaries as it cannot be certain enough to constrain the posterior in this region.

Finally, these boundary tables give us justification for ensembling when preparing for real telescope observations. For all scenarios and redshift bins, the 5-ensemble models widen the credible intervals compared to their single-model version. In the High z_3 scenario specifically, ensembling widens the boundaries by 14% for the joint model, 23% for Betti, and 32% for PS2D (compared to the single-models). Averaging slightly different normalizing flows provides a margin against overconfidence, which will be preferable when using these pipelines on real and noisy SKA observations.

When we look at why all our models struggle with the high scenario (slow reionization), we have to think about how we interpret the observation-space errors in our posterior predictive checks. In our dataset, only a small fraction of simulated parameters retain a high neutral hydrogen fraction x_{HI} through the latest redshift bin at z_3. Because we do not have access to the 21cmFAST simulator during inference, we rely on a KNN to fetch proxy observations (x_{pp}) from our training set based on our inferred parameters (θ_{pred}). For all our evaluations, the nearest-neighbour distance for the high scenario is very small, and often smaller than for the low and mid scenarios. We would normally expect that finding close parameter neighbours means the corresponding observations match the ground truth closely, but a small nearest-neighbour distance only means that our lookup found a parameter θ close to ours in the parameter-space, so it does not guarantee that the observable x maps back consistently in that region.

When relating these findings to the SDC3b data challenge and future SKA observations, we should explain that our models were trained and evaluated on observations generated by 21cmFAST. Both TARP and the PPC metrics are computed within the dataset provided by the simulator, so a strong TARP p-value only shows that our posteriors are well-calibrated with respect to 21cmFAST's algorithms. Since our scoring and calibration results are based on the 1000 hours observation scenario, we cannot say if models maintain this level of calibration under shorter integration times which are noisier such as the 100 hour observation scenario, which will be relevant when the SKA becomes operational and the first observations are evaluated.

To build a better pipeline for real SKA observations, we could base our test scenarios on observational constraints from JWST and Lyman-α measurements (Mason et al. 2026) to get a more realistic timeline of the EoR instead of defining our low, mid, and high trajectories based on the dataset percentiles ($\pm 1\sigma$ of the generated θ distribution). On the data processing side, we could explore different ways of structuring our frequency slices before passing them to the embedding networks, which would allow the models to extract more localized features across the redshift range. Once those representations are learned, we could also apply the Frequentist-Bayes (FreB) protocol to our inference outputs. Because generative models can produce overconfident or biased posteriors in complex scenarios (for example during slow reionization), FreB would reshape our output posterior distributions into locally valid confidence regions. This is a post-processing step and could guarantee that our inferred boundaries cover the true parameters even for the high scenario, where the observation-space struggles quite a bit. Refer to the paper by Carzon et al. 2026 for more information.

Chapter 7

Conclusion

In this project, we explored the use of Simulation-based Inference (SBI) to infer the parameters of the Epoch of Reionization (EoR) from mock hydrogen 21-cm observations. Specifically, we developed, trained and evaluated six architectures within the context of the SKAO SDC3b challenge: single-model and 5-ensemble pipelines for 2D power spectra (PS2D), Betti numbers, and a joint configuration combining both data types.

7.1 Summary of findings

Our evaluation showed behavioural differences between PS2D and Betti data summaries:

When comparing Betti numbers and PS2D, Betti numbers consistently achieved almost perfect parameter-space calibration (TARP p$\approx$ 1.000) and showed excellent constraining power at the fast reionization scenario in the latest redshift bin (low z_3), where credible intervals were more than seven times narrower than PS2D's. PS2D, meanwhile, netted good overall R^2 scores ($>$ 99%), but was more sensitive to architectural choices such as batch sizing, and exhibited a systematic S-shaped over/underconfidence bias across all 30 independent training trials.

In terms of the stabilizing effect of joint representations, by using a joint embedding that processes PS2D through a frozen ResNet and Betti curves through a jointly trained MLP, we found that combining both data types (which have differing structures) gives the best overall inference of the three data types. The joint model's behaviour is much closer to Betti than to a simple average of the two: at low z_3, its credible width is only 50% wider than Betti's own, still more constrained than PS2D's; at high z_3, it is about half of PS2D's width, close to Betti's own value there. PS2D's contribution mainly shows up as a small widening of Betti's already constrained posteriors, rather than the joint model pulling evenly between the two, which could be explained by the data statistics from the joint embedding being predominantly composed of Betti summaries when passing through the final layer (16 for Betti vs 3 dimensions for PS2D).

There is a trade-off in ensembling. Our 30-run trial analysis showed that ensembling only works well when the posteriors making up the models don't all share the same miscalibration. When they do (as with PS2D's S-curve, present in all 30 trials) ensembling compounds the shared miscalibration rather than averaging it out, and global TARP calibration degrades by a lot (p=0.951 to 0.739). At the same time, ensembling widened credible boundaries in every single scenario and redshift bin we checked, for every data type (at high z_3 specifically, by 14% for the joint model, 23% for Betti, and 32% for PS2D). Important thing to note: this wider uncertainty did not come at the cost of point-estimate accuracy: R^2, RMSE and MAE stayed unchanged for PS2D under ensembling, and improved slightly for both Betti and the joint model. This makes ensembling a good way to protect against overconfidence before using them on real telescope noise.

When talking about data sparsity and observation-space limits, by using parameter-space TARP diagnostics with k-nearest-neighbour Posterior Predictive Checks (PPC), we found that all models had problems mapping observables back in the slow reionization scenario (high z_3). Simulated trajectories that keep a high neutral hydrogen fraction ($x_{\mathrm{HI}} > 0.70$) through the late redshift bins are underrepresented in our dataset. The nearest-neighbour distance for this scenario was always small

across every model, often the smallest of the three scenarios, which is the opposite of what one would expect since the region is sparse.

7.2 Recommendations for the SKA

If our inference pipelines were to be used on real observations once the Square Kilometre Array (SKA) comes online, we propose using the models as follows:

1. For reliability and generalization, we recommend the joint 5-ensemble model. Real telescope observations will carry instrumental and foreground noise that 21cmFAST's mock data may not fully capture, and a single posterior has a higher risk of misspecification in that case. The joint ensemble has the best global predictive accuracy of all six configurations ($R^2 = 99.53\%$, RMSE $= 15.41 \times 10^{-3}$), a 14% wider uncertainty margin than its single-model counterpart, and still keeps a strong TARP calibration (p=0.961).

2. For rapid inference, if sampling five separate normalizing flows isn't practical given compute or storage limits, the single joint model is a good alternative. It reaches almost perfect global calibration (p=0.998) while only needing a fifth of the sampling cost, although be aware that the model has slightly tighter confidence boundaries.

It is worth noting that our pipelines were trained and evaluated only on 1000 hour integration scenarios, so these recommendations are there when assuming high signal-to-noise. If early SKA operations rely on short integration times such as 100 hours, the models would need to be re-evaluated to check whether the flows remain stable, although that can simply be done by swapping the filename in the dataloader to the 100 hour integration version of both observations and relaunching the pipelines for all 6 models.

7.3 Mandate fulfilment

Looking back at our cahier des charges (in Appendix B), we fulfilled all primary and secondary objectives. We learned the theoretical minimum needed to interpret the 21-cm signals and the cosmological redshift bins. We reviewed the state-of-the-art literature from the SDC3b challenge, implemented a reproducible MLOps training pipeline integrated with MLflow, and evaluated our models based on their precision and calibration (one thing to note however, is that we could not evaluate against the official scoring, as we could not get in contact with the people working at SKAO).

7.4 Future perspectives

While our implementation performed relatively well within the scope of a Bachelor's thesis, several things can be considered to improve upon this work.

Post-processing with local calibration Because models can be overconfident in complex scenarios such as slow reionization at z_3, future pipelines could incorporate the Frequentist-Bayes (FreB) protocol Carzon et al. 2026. As a post-processing step on our normalizing flow outputs, FreB reshapes the predicted posteriors into locally valid confidence regions, giving formal statistical coverage guarantees for extreme astrophysical trajectories without needing to retrain the model. And this goes well with the next recommendation.

Test trajectories based on empirical observations Instead of defining our evaluation scenarios (low, mid, high) purely from dataset percentiles ($\pm 1\sigma$ of the generated θ distribution), future testing could set these trajectories to empirical constraints from recent JWST and Lyman-α measurements Mason et al. 2026. This would give a more realistic benchmark of how the models perform against the true universe, as the scenarios will therefore be based on realistic constraints.

Data augmentation for late epochs To help with the uncertainty widening and PPC degradation seen in the slow reionization regime, future pipelines could address the dataset's imbalance directly through data augmentation, for example, generating more 21cmFAST simulation runs specifically to oversample high-x_{HI} trajectories at the late redshift bins. This would give the density estimator more training in a region of parameter space it currently has too little knowledge of.

Dual-space model selection Our 30-trial selection pipeline currently ranks the best model by parameter-space TARP mean p-value alone. Since we found that strong TARP calibration doesn't guarantee good observation-space (Betti's PPC degradation under ensembling being one example), future work should look at including the PPC to have a dual-criterion part of model selection together with TARP calibration to ensure the chosen best model is reliable in both spaces at once.

Integration of 3D Wavelet Scattering Transforms (WST) During our literature review, we looked at WST as an alternative compressor for non-Gaussian structure Zhao et al. 2024. We excluded it from our scope due to time constraints and the lack of raw 3D tomographic lightcone FITS cubes in our dataset. Because WST uses fixed mathematical kernels that are resilient against instrumental noise without needing supervised pretraining, integrating 3D WST summaries alongside PS2D and Betti numbers is a future next step for extracting more physical information directly from raw telescope data cubes.

Bibliography

CARZON, James et al. (Jan. 2026). *Trustworthy scientific inference with generative models*. arXiv:2508.02602 [stat.ML]. DOI: 10.48550/arXiv.2508.02602. [Accessed July 7, 2026].

CERARDI, Nicolas et al. (Nov. 2025). *Implicit inference of the reionization history with higher-order statistics of the 21-cm signal*. arXiv:2511.11568 [astro-ph]. DOI: 10.48550/arXiv.2511.11568. [Accessed Mar. 20, 2026].

COOPER, Nadia et al. (Apr. 2026). "Simulation based inference of the ionization history from the 2D 21 cm power spectrum". In: *Monthly Notices of the Royal Astronomical Society* 548.2. arXiv:2508.16329 [astro-ph], stag577. DOI: 10.1093/mnras/stag577. [Accessed May 11, 2026].

FURLANETTO, Steven, S. Peng OH, and Frank BRIGGS (Oct. 2006). "Cosmology at Low Frequencies: The 21 cm Transition and the High-Redshift Universe". In: *Physics Reports* 433.4-6. arXiv:astro-ph/0608032, pp. 181–301. DOI: 10.1016/j.physrep.2006.08.002. [Accessed Feb. 19, 2026].

LOEB, Abraham and Steven R. FURLANETTO (2013). *The first galaxies in the universe*. eng. Princeton series in astrophysics. Princeton: Princeton university press.

MASON, Charlotte A. et al. (Jan. 2026). "Constraints on the z 613 intergalactic medium from JWST spectroscopy of Lyman-alpha damping wings in galaxies". In: *Astronomy & Astrophysics* 705, A114. DOI: 10.1051/0004-6361/202553820. [Accessed July 7, 2026].

MESINGER, Andrei, ed. (2020). *The cosmic 21-cm revolution: charting the first billion years of our universe*. eng. Version: 20191201. AAS-IOP astronomy. [release 2]. Bristol, UK: IOP Publishing. DOI: 10.1088/2514-3433/ab4a73.

PIETSCHKE, Yannic et al. (Oct. 2025). "Direct reconstruction of the Reionization history from 21cm 2D Power Spectra". In: *Journal of Cosmology and Astroparticle Physics* 2025.10. arXiv:2506.19925 [astro-ph.CO], p. 039. DOI: 10.1088/1475-7516/2025/10/039. [Accessed Apr. 12, 2026].

PYTORCH CONTRIBUTORS (2026). *Reproducibility — PyTorch 2.13 Documentation [online]*. Available from: https://docs.pytorch.org/docs/2.13/notes/randomness.html [accessed 22 July 2026].

SALIS, Emmanuel de et al. (2026). "Exploring the Early Universe with Deep Learning". In: vol. 16121. arXiv:2509.22018 [astro-ph], pp. 426–438. DOI: 10.1007/978-3-032-05176-9_33. [Accessed Mar. 31, 2026].

TEJERO-CANTERO, Álvaro et al. (2026). *sbi: Simulation-based inference in Python [online]*. Available from: https://sbi.readthedocs.io/en/stable/ [accessed 22 July 2026].

ZHAO, Xiaosheng et al. (Sept. 2024). *Simulation-based Inference of Reionization Parameters from 3D Tomographic 21 cm Light-cone Images – II: Application of Solid Harmonic Wavelet Scattering Transform*. arXiv:2310.17602 [astro-ph]. DOI: 10.48550/arXiv.2310.17602. [Accessed Apr. 21, 2026].